DETERMINISTIC FACTORING OF BALANCED SEMIPRIMES IN AVERAGE TIME $N^{1/6}(\log N)^{O(1)}$

Abstract. We give a deterministic algorithm that receives only an integer N and outputs its complete prime factorisation. On every input its cost is that of the Harvey–Hittmeir algorithm, $O(N^{1/5}\log^{16/5}N/(\log\log N)^{3/5})$ bit operations, up to lower-order terms. Averaged over the uniform distribution on the n-bit semiprimes $N = pq$ with primes $p < q \le 2p$, its cost is $N^{1/6}(\log N)^{O(1)}$. We know of no previously proved average-case or almost-all deterministic exponent below $1/5$ for this population.

The engine is a deterministic weighted collision search, which is a Lehman-type method. It takes the McKee–Pinch divisibility event $\ell \mid Bp - Aq$ over all primes $\ell \equiv 3 \pmod 4$ in four dyadic bands near $N^{1/6}$ and all small coprime weights (A, B). It adds the Harvey–Hittmeir modulus-squared lock, and one baby-step list per band, shared by all primes of the band through cofactor-powered handles. One run with weight height H costs $N^{1/6}H^{5/3}(\log N)^{O(1)}$ on every input and never outputs a wrong answer. Its coverage step misses the true factor on at most an $O((\log N)/H^2)$ and an $O((\log N)^4/H^4)$ fraction of the pairs; the second bound comes from a two-variable fourth-moment estimate. It gives, for every $\varepsilon \le 1/40$, cost $N^{1/6+5\varepsilon/12+o(1)}$ with exceptional fraction $O(N^{-\varepsilon})$. The complete algorithm reruns the search with heights $H \leftarrow H^{6/5}$ behind a baby-step giant-step detector for primes with small $\mathrm{ord}_q(2)$. This works because the proved failure fraction decays faster in H than the cost grows.

We prove two further results. First, the worst case reduces to coverage alone: a modified search factors every balanced N whose true pair is caught by one band. So a pair-covering hypothesis implies worst-case cost $N^{1/6+o(1)}$, and any covering theorem with weight height at most $Y^{1/2}J^{-c}$, for some $c > 3/8$, would beat $1/5$ in the worst case. Second, in a sumset collision-search cost model with residue-local, degree-one targets, reaching the true factor of a constant fraction of the pairs requires cost $N^{1/6}/(\log N)^{O(1)}$. Our search attains this up to polylogarithmic factors. (Here costs are per input, or averaged over all n-bit integers; budgets averaged only over the semiprimes are not covered.) The result says nothing about other kinds of algorithm, and our cost proxy at 40–80 bits favours Harvey–Hittmeir.

DRAFT — not for distribution. This draft was written with substantial AI assistance (see the Acknowledgements). Its proofs have been checked by AI referees and have not yet been checked by a human referee. Status labels are used throughout: *proved*, *experimentally supported*, *conjectured* and UNVERIFIED.

Contents

1. Introduction and statement of results 2
1.1. Background 2
1.2. Population and cost 3
1.3. Main results 3
1.4. How it works 5
1.5. What is new, and what is not 5
1.6. Organisation 6
2. Prior art 6
3. The search $\mathsf{DCL}_{n,H}$ 9
3.1. Parameters 9
3.2. Steps 9

Date: DRAFT — not for distribution. Version of 18 October 2026.

4. Analysis of one run 10
4.1. Lock, coverage and windows 10
4.2. Exact collision modulo p; the clean band 10
4.3. Counting lemmas 11
4.4. One run 12
4.5. Bit complexity 13
5. A fourth-moment coverage bound 13
5.1. Setting 13
5.2. Consequences 15
5.3. Where this method stops 16
6. A complete algorithm with average cost $N^{1/6}$ 16
6.1. The small-order detector 16
6.2. The algorithm LAD 17
6.3. Proof of Theorem 1.1 17
7. The worst case reduces to coverage 18
7.1. Setting 18
7.2. The algorithm DCL-wc 18
7.3. Theorem R 19
7.4. Consequences 20
8. A true-factor floor for local designs 22
8.1. The model 22
8.2. The bound 22
9. Unbalanced ratios 24
9.1. Ratio bands 24
9.2. The envelope 25
9.3. A full-coverage barrier 25
9.4. Which populations the results serve 26
10. Experiments 27
10.1. The search DCL (pre-registration `05d2bf5`) 27
10.2. Coverage statistics (pre-registration of `sieve2d`) 27
10.3. Worst-case coverage and Theorem 7.1 (pre-registration of `dchworst`) 28
11. Limitations and open problems 28
Acknowledgements 28
Appendix A. Full-candidate covering bounds (context; statements only) 29
Appendix B. Reproducibility 30
References 30

1. Introduction and statement of results

1.1. **Background.** Few exponents are known for deterministic integer factoring with rigorous bounds.

- Trial division costs $N^{1/2}$.
- Lehman's method [17] gives $N^{1/3+o(1)}$.
- The Pollard–Strassen method [23, 25] gives $N^{1/4+o(1)}$. It was refined by Bostan–Gaudry–Schost [2] and Costa–Harvey [5].
- Hittmeir saved a subexponential factor at exponent 1/4 [14]. He then proved exponent 2/9 with a time–space trade-off for Lehman's method [15].

- Harvey [9] proved exponent 1/5. Harvey and Hittmeir [10] improved the logarithmic factor, and Gao, Feng, Hu and Pan [8] improved it again for balanced semiprimes, at the same exponent.

Below 1/5 the known routes are conditional or partial.

- Umans and Wang [26] showed that a divisor-cover conjecture would give 1/6. He and Sahai [13] refuted that conjecture's one-dimensional arithmetic-progression form.
- Pollard's $p-1$ method succeeds on a positive proportion of the nearly balanced semiprimes pq with $x^{1-\varepsilon} < p, q \le x$ at exponents approaching $1/(4\sqrt{e}) \approx 0.1516$. This is Ward–Maestro [28, Prop. 35.3], deduced from Friedlander's theorem on smooth shifted primes [7] and Fouvry's on shifted primes with a large prime factor. They note that it gives no average-time bound for a complete algorithm: the failure set has positive density.
- Carella [4] claims a deterministic $O(N^{1/6+\varepsilon})$ algorithm. Its displayed proof has a normalisation defect (Section 2).

This paper is about the *average* cost over the balanced population, the semiprimes with two factors of comparable size. For unrestricted semiprimes almost all have a factor below N^{δ} [6], so the typical input is easy, although an average over a complete algorithm can still be dominated by a sparse expensive tail. Balanced semiprimes are the opposite. Every method above is analysed in the worst case or on a positive proportion, and we are not aware of a proved average-case or almost-all deterministic exponent below 1/5 for them.

1.2. **Population and cost.**

Let n be the bit length, $P = 2^{n/2}$, and

$$\mathcal{S}_n = \{(p,q) : p < q \le 2p,\ p,q \text{ prime},\ 2^{n-1} \le pq < 2^n\}.$$

We count $\mathcal{S}_n$ by pairs. By unique factorisation this is the same as counting the semiprimes $N = pq$, and "uniform on $\mathcal{S}_n$" means uniform on these semiprimes. For $(p,q) \in \mathcal{S}_n$ we have $p \in [\sqrt{N/2}, \sqrt{N}] \subset [P/2, P]$ and $q \in (P/\sqrt{2}, \sqrt{2}P)$. The prime number theorem gives $c_0 P^2/n^2 \le |\mathcal{S}_n|$ for an absolute $c_0 > 0$ and all large n. Costs are bit operations on a multitape Turing machine, with integer multiplication in time $M(b) = O(b \log b)$ [12]; $\tilde{O}(\cdot)$ hides factors polynomial in n, and a *word operation* is an operation on $O(n)$-bit integers. $\mathrm{HH}(N) = O(N^{1/5} \log^{16/5} N/(\log\log N)^{3/5})$ denotes the Harvey–Hittmeir bound [10].

1.3. **Main results.**

Theorem 1.1 (average-case complete factoring; proved)**.** *There is a deterministic algorithm* LAD *and an absolute constant c with the following properties.*

(i) Completeness. *On every input $N \ge 2$,* LAD *outputs the complete prime factorisation of N.*

(ii) Worst case. *On every n-bit input its cost is at most*

$$\mathrm{HH}(N) + O(n^{c+4} 2^{7n/36}) + O(n^{O(1)} 2^{3n/20}) = O(\mathrm{HH}(N)).$$

(iii) Average over $\mathcal{S}_n$. *For all large n,*

$$\frac{1}{|\mathcal{S}_n|} \sum_{(p,q)\in\mathcal{S}_n} \mathrm{cost}(pq) = O\big(2^{n/6} n^{c+1} \log n\big),$$

(iv) Other distributions. *The bound in (iii), multiplied by K, holds for every distribution on $\mathcal{S}_n$ with point masses at most $K/|\mathcal{S}_n|$.*

that is, $N^{1/6}(\log N)^{c+1}\log\log N$ on average. The constant c is the polylogarithmic exponent in the cost of one run of the search (Proposition 4.13(ii)). Item (iv) covers, for instance, the product of two independent uniform $n/2$-bit primes conditioned to be distinct, for even n: its support lies in $\mathcal{S}_n \cup \mathcal{S}_{n-1}$ and its mass ratio is bounded (apply (iii) to each shell). It says nothing about uniform n-bit integers, or about key-generation rules that restrict p and q further.

Theorem 1.2 (almost-all factoring at fixed cost; proved)**.** *There are deterministic algorithms $\mathsf{DCL}_{n,H}$ (Section 3) with the following property. Each sees only N. Each outputs either a proper divisor of N or* `FAIL`*, and never anything else.*

- *(a) For every fixed $\varepsilon \in (0, 1/40]$ and $n \geq n_0(\varepsilon)$, the choice $H = \lceil n2^{\varepsilon n/4} \rceil$ gives cost $N^{1/6+5\varepsilon/12}(\log N)^{O(1)}$ on every n-bit input. It outputs* `FAIL` *on $O(2^{-\varepsilon n})|\mathcal{S}_n|$ pairs of $\mathcal{S}_n$, i.e. on a fraction $O(N^{-\varepsilon})$.*
- *(b) For every fixed $a > 0$, the choice $H = \lceil n^{1+a} \rceil$ gives cost $N^{1/6}(\log N)^{O(1+a)}$. It outputs* `FAIL` *on $O\big(n^{-\max(4a,\,1+2a)}\big)|\mathcal{S}_n|$ pairs.*

An earlier version of (a), proved with the arithmetic large sieve alone, gave cost $N^{1/6+5\varepsilon/6+o(1)}$ for $\varepsilon \leq 1/60$ (Theorem D′ of the project notes; Remark 4.14). The improvement comes from the two-variable fourth moment of Section 5.

Theorem 1.3 (worst case from coverage; proved)**.** *There is a deterministic algorithm DCL-wc (Section 7) that takes as public parameters a weight set and blocks of primes. Suppose $N = pq \in \mathcal{S}_n$ and some block contains a prime ℓ with $\ell \mid Bp - Aq$ for some weight (A, B). Then DCL-wc outputs a proper divisor of N. Its cost is $N^{1/6+o(1)}$ plus the list cost of the shared collision search over the blocks, including target construction and rejected (weight, prime) tests (Theorem 7.1(iii)). No hypothesis on any multiplicative order is needed.*

Corollary 1.4 (proved)**.** *(a) If Hypothesis 7.5 ($\mathrm{DCH}_{\mathrm{pair}}(\kappa)$) holds for some κ, then every $N \in \mathcal{S}_n$ is factored deterministically in $N^{1/6+o(1)}$ bit operations.*

- *(b) Fix $c \in (3/8, 1/2]$. Suppose that for all large n there is an n-only band scale $Y = N^{1/(7-2c)+o(1)}$ such that every pair of $\mathcal{S}_n$ has a prime $\ell \equiv 3 \pmod 4$ in $(Y, 2Y]$ dividing some $Bp - Aq$, with a primitive weight of height $H \leq Y^{1/2}J^{-c}$, where $J \asymp Y/\log Y$ is the number of such primes. Then every $N \in \mathcal{S}_n$ is factored deterministically in $N^{(1-c)/(7/2-c)+o(1)}$ bit operations, and $(1-c)/(7/2-c) < 1/5$.*

Theorem 1.5 (true-factor floor for local designs; proved in the model)**.** *Consider a sumset collision search in which labels are charged, at most X targets are issued and the windows have total size at most X^2. Suppose its targets come from a fixed family of residue-local degree-one generators (Section 8), and that the two budgets hold per input or in expectation over all n-bit integers. Then the prescribed coverage event (a target certifies the true factor) holds for at most $Cn^2X^{3/2}2^{-n/4}|\mathcal{S}_n|$ pairs. Coverage of a constant fraction σ of $\mathcal{S}_n$ therefore needs $X \geq c\,\sigma^{2/3}2^{n/6}n^{-4/3}$. The theorem does not bound successes through accidental collisions outside the coverage event.*

Theorem 1.2(b) attains the bound of Theorem 1.5 up to polylogarithmic factors: the algorithm $\mathsf{DCL}_{n,H}$ is dominated, at constant-factor cost, by a design of this kind (Proposition 8.3). This replaces a claim made in an earlier draft that the algorithm is "optimal in its model". That claim was based on a lower bound for covering *every* integer candidate (Appendix A), which does not bound an algorithm that only needs to reach the true factor.

Remark 1.6 (scope)**.** Each item below is easy to over-read, so we state them plainly.

- (a) Theorems 1.1 and 1.2 concern inputs drawn from $\mathcal{S}_n$, with exceptions counted by pairs. Neither gives a worst-case bound below $1/5$. Theorem 1.1 is an *expected-time* statement

for a complete algorithm. By Markov's inequality it implies tail bounds on the cost; Theorem 1.2 supplies sharper explicit tails in its parameter range.

(b) Theorems 1.1 and 1.2 treat only ratios $q/p \le 2$. Section 9 extends the almost-all statement to $q/p \le N^{o(1)}$. It also shows that our methods leave a ratio gap peaking at $N^{1/5}$ at $q/p \approx N^{1/5}$, and Proposition 9.3 shows that full-coverage searches in a Lehman-linear model cannot close it.

(c) "Never wrong" means that every output is validated.

(d) Theorem 1.5 is a theorem inside a stated cost model. It is not a lower bound for factoring. Pollard's $p-1$ method, which lies outside the model, already reaches a positive proportion of a nearly balanced family below $N^{1/6}$ ([28, Prop. 35.3], [7]). The cost budget in Theorem 1.5 is per input or averaged over all n-bit integers; budgets averaged only over $\mathcal{S}_n$ are not covered.

(e) All statements are asymptotic. At 40–80 bits the search is 2.3–4.9 times more expensive than Harvey–Hittmeir in our cost proxy (Section 10).

1.4. **How it works.** Fix a prime ℓ and coprime weights (A, B).

- If $\ell \mid Bp - Aq$, then $p \bmod \ell$ is one of the two square roots of AN/B modulo ℓ, and these roots are computable from N.
- On that residue class the weighted sum $Bp + Aq$ is determined modulo ℓ^2 (the Harvey–Hittmeir lock). So $g = (Bp + Aq - c)/\ell^2$ is an integer in a window of length about $H\sqrt{N}/\ell^2$.
- Exponentiating, $2^{(L/\ell^2)(B+AN-c)} \equiv \gamma^g \pmod p$, with $\gamma = 2^L$ and $L = \prod \ell^2$ over the band. So every target, for every prime of the band, is one giant step against the band's single baby list $\gamma^0, \dots, \gamma^{m-1}$.
- Primes near $N^{1/6}$ balance the baby list ($\approx H\sqrt{N}/\ell^2$) against the number of targets ($\approx H^2\ell$).
- Coverage of the true p is a sieve problem: the large sieve gives the failure fraction $O(n/H^2)$, and a fourth moment over both variables gives $O(n^4/H^4)$.
- The collision modulo p is an exact identity, so no order lemma is needed there. It remains to rule out a simultaneous collision modulo q. Four disjoint bands make one band coprime to $q-1$, and on that band a two-sided collision forces $\mathrm{ord}_q(2)$ to divide a specific small expression, which is rare.

The failure fraction falls like H^{-2} while the cost grows like $H^{5/3}$. So rerunning on failure with $H \leftarrow H^{6/5}$ costs $N^{1/6}$ polylog per rung on average, and after $O(\log\log N)$ rungs only a fraction $N^{-1/30}$ of pairs remains for the $N^{1/5}$ fallback. The order condition is handled up front by a baby-step giant-step search for the least k with $\gcd(2^k - 1, N) > 1$, up to $N^{3/10}$.

1.5. **What is new, and what is not.** We claim novelty only in the form "we are not aware of". Each ingredient is known (Section 2).

- The divisibility event $\lambda p \equiv \pm\mu q \pmod k$ with small multipliers, the N-determined residues and a modulus-squared reduction are due to McKee and Pinch [18, §5]. So is a *heuristic* balanced method over primes $k \approx N^{1/4}$.
- The modulus-squared lock in the form we use is Harvey–Hittmeir [10, Prop. 3.3].
- Shared baby-step/giant-step collision search with multipoint evaluation over $\mathbb{Z}/N\mathbb{Z}$ is Hittmeir's [14], specialised to powers and to the simultaneous Lehman search by Harvey [9, Alg. 4.1]. Exact-match handling is [9, Alg. 4.2].
- The order-selection tools behind Theorem 1.3 adapt Harvey–Hittmeir 2026 [11].
- The large sieve is Montgomery's [19, 20].

We are not aware of a prior statement of any of the following:

(i) a band of primes at $N^{1/6}H^{-1/3}$, made possible by one baby list per band shared through cofactor-powered handles;
(ii) a proof that the event catches the true factor for all but $O(n/H^2)$, or $O(n^4/H^4)$, of the pairs;
(iii) for the average case, an extraction argument with no order hypothesis: the collision modulo p is an unconditional identity, and a clean-band argument bounds the two-sided collisions on average; for the worst case, a separate extraction algorithm (Theorem 1.3);
(iv) the (routine, once the tail bound is available) observation that the proved tail beats the cost growth, which turns the almost-all algorithm into a complete one with average exponent 1/6 by standard restarts and a baby-step giant-step order search. The substantive input to Theorem 1.1 is the one-run analysis (Proposition 4.13);
(v) the reduction of the worst case to coverage (Theorem 1.3);
(vi) a true-factor lower bound for local designs (Theorem 1.5).

The full-candidate covering floor $N^{1/6}$ of Appendix A is *not* new. The same exponent, by a closely related argument, is in Ward–Maestro [28, Thm. 3.6] (2 September 2026) for Lehman forms. Their model allows a common modulus: thinning by residue classes (Lemma 2.3, at the free-giant level) and Harvey–Hittmeir-type single-class locks (Theorem 2.8, with materialised giants). Our appendix version allows per-target moduli at the free-giant level. We derived it independently and later.

1.6. **Organisation.**

- Section 2 reviews prior art.
- Section 3 defines the search $\mathsf{DCL}_{n,H}$, and Section 4 analyses one run.
- Section 5 proves the fourth-moment coverage bound and Theorem 1.2.
- Section 6 proves Theorem 1.1.
- Section 7 treats the worst case (Theorem 1.3, Corollary 1.4).
- Section 8 proves Theorem 1.5.
- Section 9 treats unbalanced ratios.
- Section 10 reports pre-registered experiments, and Section 11 lists open problems.
- Appendix A contains the full-candidate covering bounds.

2. Prior art

We read McKee–Pinch §5, Ward–Maestro and Carella §5 ourselves. The other descriptions come from GPT-6.1-sol literature sweeps (see the Acknowledgements), which checked statements against the sources. They are marked *[sweep]*. Where even the sweep did not check a source, we say UNVERIFIED.

Lehman and weighted Fermat searches. Lehman [17] searches for $4kN = x^2 - y^2$ over small multipliers k and proves a deterministic $N^{1/3+o(1)}$ bound [sweep]. Weighted Fermat searches go back to Lawrence [16] (UNVERIFIED). The weighted sums $Bp + Aq$ used here are Lehman's mechanism.

McKee–Pinch 1996, §5: the direct precursor. We read §5 (pp. 221–223) of [18] from the public scan indexed at `https://cr.yp.to/bib/1996/mckee-deterministic.html`. It contains five things.

- Lehmer's speed-up: if $p \equiv q \equiv 1 \pmod{k}$, Fermat's method runs in $O(q/k^2)$.
- An extension: for $p \equiv \pm q \pmod{k}$, Fermat runs in $O(q/k^{2-\epsilon})$, with u or v in $n = u^2 - v^2$ determined modulo k^2 from square roots modulo k. This is a modulus-squared lock.
- Multipliers: if $\lambda p \equiv \pm \mu q \pmod{k}$, the same method splits $\lambda\mu n$.
- A deterministic $O(n^{1/3+\epsilon})$ algorithm, built on a pigeonhole lemma.

- A heuristic balanced method, described by the authors as "not however an algorithm". It takes primes $k \in (n^{1/4}, 2n^{1/4})$ and small λ, μ, and hopes that $\lambda p - \mu q \equiv 0 \pmod{k}$ for some of them, at expected cost $O(n^{1/4+\epsilon})$.

So the core event of this paper is theirs: a band prime divides a small-weight combination, with N-determined residues and a modulus-squared reduction. We trade a larger window for fewer, smaller primes through one baby list per band. We also prove the coverage statement, the extraction, the average-time theorem and the worst-case reduction.

Hittmeir 2018, 2021. Hittmeir [14] combines baby-step/giant-step with sieving of $p+q$ modulo many small primes and proves $N^{1/4}\exp(-\Omega(\log N/\log\log N))$ [sweep]. Hittmeir [15] proves $N^{2/9+o(1)}$ through a time–space trade-off for Lehman's method [sweep]. Hyperbola sieving over many small primes is therefore old. What differs here is that each prime is used *selectively*: only the at most two classes reached by a short weight vector are used, rather than all classes being covered.

Harvey 2021; Harvey–Hittmeir 2022. Harvey [9] proves the deterministic bound

$$O(N^{1/5}\log^{16/5} N),$$

using a single shared baby-step giant-step search over Lehman's family. He uses the congruences

$$\alpha^{aq+bp} \equiv \alpha^{aN+b} \pmod{p}$$

and multipoint evaluation over $\mathbb{Z}/N\mathbb{Z}$ (Alg. 4.1, which Harvey describes as almost identical to Hittmeir's Algorithm 3.2 [14]), with exact matches handled in Alg. 4.2 [sweep]. Harvey–Hittmeir [10] save a factor $(\log\log N)^{3/5}$. Their Proposition 3.3 contains the congruence $aq - bp \equiv 0 \pmod{m}$ and the modulus-squared lock we use as Lemma 4.1, with a single modulus and lattice-chosen weights [sweep]. Harvey remarks that exponent $1/6$ would need a "fully square-root speedup" of Lehman's search [sweep].

Harvey–Hittmeir 2026; Oznovich–Volk 2025. These works give deterministic algorithms that find an element of large multiplicative order modulo N, or a factor [11, 21] [sweep]. Theorem 1.3 adapts Harvey–Hittmeir's smooth-number subgroup argument and their order dichotomy ([11, Lemmas 2.1, 2.3, 2.4 and §3.1] [sweep]) to a one-sided reading: the first r with $\gcd(b^r-1, N) > 1$ gives a factor unless the two orders coincide. Large order modulo N does not give large order modulo a specific prime. Our average-case theorems therefore never need an element of large order: $\mathrm{ord}_q(2)$ enters only in counting the exceptional sets (E_a, E_c, $C(T)$) and in the cheap detector of Section 6.

Ward–Maestro 2026. We read the report [28] (dated 2 September 2026, 110 pages, six parts). Three parts of it bear on this paper.

- *The covering floors.* The report defines an "oblivious Lehman-cell BSGS" model (Definition 2.1): forms $aq + bp$ with $a, b \geq 1$ and tested windows that must cover every real p in the range. In that model it proves floors of $N^{1/3}$ candidates (Corollary 3.3), $N^{1/5}$ with materialised giants (Theorem 3.4) and $N^{1/6}$ with free giants (Theorem 3.6). It also treats unbalanced ranges, forms of bounded degree (Theorem 9.4), thinning by residue classes modulo $q \leq N^{1/20}$ (Theorem 2.8), and randomised covering (Remark 3.5).
- *Partial information.* It gives deterministic searches given partial information, including $N^{1/5+o(1)}M^{-2/5}$ given $p \bmod M$ with $M \leq N^{1/7}$ (Proposition 14.2).
- *The $p-1$ result.* It proves the positive-proportion $p-1$ result quoted in the introduction (Proposition 35.3).

Its $N^{1/6}$ floor and the main idea of its proof (the curvature bound on the covering sum, Lemma 3.2) pre-date and overlap our Appendix A. Our version allows per-target lock moduli and charges window counts per modulus, but the exponent and its proof idea are theirs.

- The report contains no algorithm with an almost-all or average-case guarantee below $N^{1/5}$.
- It does not use many-prime selective covering, a prime band near $N^{1/6}$, or a baby list shared across moduli.
- It considers no true-factor counting under residue-local keys of the kind in Theorem 1.5.

Proposition 14.2 there is related in spirit to our targets, which compute $p \bmod \ell$ from the event $\ell \mid Bp - Aq$ rather than receive it. Ward–Maestro cite Carella's claim as unverified.

Stănică–Mulder–Hittmeir 2026. [24] studies unions of weighted modular-hyperbola sieve sets, their CRT behaviour and elliptic-trace obstructions [sweep]. Its averages are over finite-field parameters. The sweep found no global coverage theorem over prime pairs in it.

Gao–Feng–Hu–Pan 2025. For balanced semiprimes, [8] claims a deterministic algorithm of cost

$$O(N^{1/5} \log^{13/5} N/(\log\log N)^{3/5})$$

via rank-3 lattices [sweep]. The exponent remains 1/5.

Umans–Wang 2025; He–Sahai 2026. Umans and Wang [26] show that a Strong Prefactored Divisor Conjecture would give deterministic exponent 1/6 (their Thms. 5.2 and 5.5). He and Sahai [13] refute the one-dimensional arithmetic-progression version at the proposed parameters; higher ranks remain open [sweep]. Our results are unconditional but average-case. The covering statement that would make our method worst-case (Hypothesis 7.5) is a different object. We know of no implication between it and the Umans–Wang conjecture in either direction, and we know of no transfer of the He–Sahai argument to it.

Carella, arXiv:1308.2891. Carella [4] states (Theorem 1.1) a deterministic algorithm that factors a balanced $N = pq$ in $O(N^{1/6+\varepsilon})$ operations. The algorithm guesses the least significant $(\log N)/6$ bits of p and recovers the rest by a bivariate small-root method. We read its §5. The argument there runs as follows.

- Put $B = \lfloor N^{1/6} \rfloor$. Write $p = x_1 B^3 + Bx + x_0$ and $q = y_1 B^3 + By + y_0$ with $|x|, |y| \le N^{1/3}$, where $y_0 \equiv N x_0^{-1} \pmod B$.
- Equation (17) considers $f(x, y) = pq - N$ and lists its coefficients.
- The listed coefficients of x and y are $y_1 B^3 + y_0$ and $x_1 B^3 + x_0$, and the coefficient of xy is B^2.
- From this list f is declared primitive. With $X = Y = N^{1/3}$ its height is $W = N$, and the small-root condition $XY < W^{2/3}$ is invoked.

Expanding the product directly gives linear coefficients $B(y_1 B^3 + y_0)$ and $B(x_1 B^3 + x_0)$. The constant term $(x_1 B^3 + x_0)(y_1 B^3 + y_0) - N$ is also divisible by B, because $x_0 y_0 \equiv N \pmod B$. So f has content divisible by B. The small-root theorem applies to the primitive polynomial f/B, whose height is about $N/B = N^{5/6}$. For it, $XY = N^{2/3}$ exceeds $(N^{5/6})^{2/3} = N^{5/9}$, so the condition fails. (Even before this correction the condition holds only with equality.) This matches the familiar fact that small-root methods need about $(\log N)/4$ known bits of a factor, not $(\log N)/6$. We conclude only that the displayed argument does not establish the claimed bound. The same defect was found independently in three of our AI reviews.

Lower bounds in restricted models. Lower bounds for factoring exist only in restricted models. Brown [3] bounds the length of straight-line factoring programs, a class that includes a variant of the elliptic curve method. Ward–Maestro prove floors in covering models. Theorem 1.5 is of the same kind, a statement inside a stated cost model, and we make no claim about factoring lower bounds in general.

The large sieve; moments. We use the arithmetic large sieve in Montgomery's form for squarefree moduli [19], with the Montgomery–Vaughan constant [20]. The barrier remark in Section 5 uses Peled–Yadin–Yehudayoff [22].

Summary. Our own searches, and GPT-6.1-sol sweeps checking about twenty items, found no unconditional, oracle-free deterministic result below exponent 1/5 that factors almost all, or on average, balanced prime pairs. The only exception is Carella's claim above. This is the outcome of a search, not a proof of priority.

3. The search $\mathrm{DCL}_{n,H}$

3.1. **Parameters.** All parameters depend on n and on an integer weight height H, never on N. We assume throughout

$$H_0 \le H \le 2^{n/50}, \qquad n \ge n_0, \tag{1}$$

for absolute constants H_0, n_0. The constant n_0 is an eventual, very large guard: for example, the top rung of Section 6 needs $n^2 2^{\lceil n/60 \rceil} \le 2^{n/50}$, which holds only for n in the thousands. The experiments of Section 10 run far below it.

- The weights are $V = \{(A,B) \in [1,H]^2 : \gcd(A,B) = 1,\ A \ne B\}$. For $H \ge H_0$ we have $H^2/2 \le |V| \le H^2$.
- The band scale is $Y = \lceil 2^{n/6} H^{-1/3} \rceil$.
- There are four bands, $k = 0,1,2,3$:

$$\mathcal{P}_k = \{\ell \text{ prime} :\ \ell \equiv 3 \pmod 4,\ 2^k Y < \ell \le 2^{k+1} Y\}, \qquad L_k = \prod_{\ell \in \mathcal{P}_k} \ell^2.$$

- The window bound is $\overline{\mathrm{span}} = \lfloor \frac{5}{2} H(2^{n/2}+1) \rfloor + 2H + 3$, and $\bar{m}_k = \lfloor \overline{\mathrm{span}}/(2^k Y)^2 \rfloor + 2$.
- The base is $\beta = 2$.

Under (1) the following scale separations hold:

$$Y \ge 16H^2, \qquad 16Y < P/2, \qquad Y^4 > \sqrt{2}\,P, \qquad H\lceil \sqrt{2} P \rceil < Y^4. \tag{2}$$

Indeed $H^{7/3} \le 2^{7n/150} < 2^{n/6}/32$, and $Y^4 \ge 2^{(2/3-4/150)n} > 2^{n/2+1} H$.

3.2. **Steps.** Input: N with n bits.

1. If $N < 4$ output `FAIL`. If N is even output 2. Put $a_0 = \lceil \sqrt{N/2} \rceil$ and $b_0 = \lfloor \sqrt{N} \rfloor$. Every $(p,q) \in \mathcal{S}_n$ has $p \in [a_0, b_0]$.
2. For $v = (A,B) \in V$ let

$$F_v(x) = Bx + AN/x, \qquad \mathrm{Lo}_v = \lfloor 2\sqrt{ABN} \rfloor, \qquad \mathrm{Hi}_v = \max_{x \in \{a_0, b_0\}} \lceil F_v(x) \rceil.$$

   - By AM–GM, $\mathrm{Lo}_v \le F_v(x)$ for all $x > 0$.
   - By convexity, $F_v \le \mathrm{Hi}_v$ on $[a_0, b_0]$.
   - $\mathrm{Hi}_v - \mathrm{Lo}_v \le (B + \sqrt{2}A)\sqrt{N} + 2H + 2 \le \overline{\mathrm{span}}$.

Then, for each block $k = 0,1,2,3$:

3. *Exponent tree.* Build the balanced integer product tree of $\{\ell^2 : \ell \in \mathcal{P}_k\}$, and push 2 down the tree: a node holding $2^{L_k/\Pi(\text{node})}$ passes $2^{L_k/\Pi(\text{node})\cdot\Pi(\text{sibling})}$ to each child, where Π denotes the product of the leaves below a node. This yields $u_\ell = 2^{L_k/\ell^2} \bmod N$ for all $\ell \in \mathcal{P}_k$. Put $\gamma = u_{\ell_0}^{\ell_0^2} = 2^{L_k} \bmod N$. If some $\ell \in \mathcal{P}_k$ divides N and $1 < \ell < N$, output ℓ.
4. *Targets.* For $\ell \in \mathcal{P}_k$ and $v \in V$, let $a = ANB^{-1} \bmod \ell$. If $a^{(\ell-1)/2} \not\equiv 1 \pmod \ell$, skip. Otherwise put $r_0 = a^{(\ell+1)/4} \bmod \ell$; the roots of $Br^2 \equiv AN \pmod \ell$ are r_0 and $\ell - r_0$. For each root r put

$$s = Nr^{-1} \bmod \ell^2, \quad c = (Br + As) \bmod \ell^2, \quad g_{\mathrm{lo}} = \left\lceil \frac{\mathrm{Lo}_v - c}{\ell^2} \right\rceil, \quad g_{\mathrm{hi}} = \left\lfloor \frac{\mathrm{Hi}_v - c}{\ell^2} \right\rfloor,$$

and $W_\tau = g_{\mathrm{hi}} - g_{\mathrm{lo}} + 1 \le \bar{m}_k$. The triple $\tau = (v, \ell, r)$ is a *target*. Let $T_k \le 2|V||\mathcal{P}_k|$ be the number of targets. If $T_k = 0$, skip the block.

5. *Baby length.* Choose m either as $m = \bar{m}_k$, or by the *endpoint-competitive* rule: minimise $m + \sum_\tau \lceil W_\tau/m \rceil$ over an integer geometric grid whose last point is $\max_\tau W_\tau$. Under either rule $m \le \bar{m}_k$ and $m + T' \le \bar{m}_k + T_k$, where $T' = \sum_\tau \lceil W_\tau/m \rceil$.
6. *Giants.* Precompute $w_\ell = u_\ell^N$. For each target and $0 \le j < \lceil W_\tau/m \rceil$,
$$G_{\tau,j} = w_\ell^A \, u_\ell^{B-c+\ell^2} \, \gamma^{-(g_{\mathrm{lo}}+1)} \, \gamma^{-jm} \equiv u_\ell^{B+AN-c} \, \gamma^{-(g_{\mathrm{lo}}+jm)} \pmod N.$$
7. *Collision detection.* Build $f(x) = \prod_{0 \le i < m} (x - \gamma^i) \in (\mathbb{Z}/N\mathbb{Z})[x]$ by a product tree, and evaluate f at all giants by a remainder tree. Only monic divisors occur, so this is valid over $\mathbb{Z}/N\mathbb{Z}$. For each giant compute $\gcd(f(G), N)$, and output it if it lies strictly between 1 and N.

If no block outputs a divisor, output `FAIL`.

Remark 3.1. (a) No multiplicative order is computed, assumed or certified, and no randomness is used.

(b) The weight $A = B$ is excluded. By Lemma 4.5 its true giant always collides modulo q too.

(c) One of the two rules in step 5 is mandatory for the cost bound. The bound $m \le \bar{m}_k$ alone would allow $m = 1$, making T' too large.

(d) An implementation (Python, gmpy2, python-flint) is `src/dcltheorem/algorithm.py`. Every output x is checked for $1 < x < N$ and $x \mid N$. An exhaustive regression covers all $2 \le N \le 10^4$.

4. Analysis of one run

Throughout this section fix $(p, q) \in \mathcal{S}_n$, $N = pq$ and a block k. Write $L = L_k$, $u = u_\ell$ and $d = \mathrm{ord}_q(2)$. By (2) every band prime is smaller than both factors and exceeds H.

4.1. Lock, coverage and windows.

Lemma 4.1 (modulus-squared lock; [10, Prop. 3.3])**.** *Let $\ell \in \mathcal{P}_k$, $(A, B) \in V$, and let r be a root of $Br^2 \equiv AN \pmod \ell$. If $p \equiv r \pmod \ell$, then $g := (Bp + Aq - c)/\ell^2 \in \mathbb{Z}$.*

Proof. $\ell \nmid ABN$. Write $p = r + \ell x$. Modulo ℓ^2, $p^{-1} \equiv r^{-1}(1 - \ell x r^{-1})$, so $q \equiv s - \ell x \, sr^{-1}$ and $Bp + Aq \equiv c + \ell x(B - Asr^{-1})$. Since $sr^{-1} \equiv Nr^{-2}$ and $Br^2 \equiv AN$ modulo ℓ, the last term vanishes modulo ℓ^2. □

Lemma 4.2 (coverage criterion)**.** *$p \bmod \ell$ is one of the computed roots if and only if $\ell \mid Bp - Aq$. In that case the target $\tau = (v, \ell, p \bmod \ell)$ exists; we say that τ* covers *p.*

Proof. $Bp^2 - AN = p(Bp - Aq)$ and $\ell \nmid p$. If $\ell \mid Bp - Aq$ then $a \equiv p^2$ is a nonzero square, and for $\ell \equiv 3 \pmod 4$ its roots are $\pm a^{(\ell+1)/4}$. □

Lemma 4.3 (window)**.** *If τ covers p, then $g_{\mathrm{lo}} \le g \le g_{\mathrm{hi}}$. Put $i_{\mathrm{full}} = g - g_{\mathrm{lo}}$, $j^* = \lfloor i_{\mathrm{full}}/m \rfloor$ and $i^* = i_{\mathrm{full}} - j^* m$. Then G_{τ,j^*} is computed; we call it the* true giant*.*

Proof. $\mathrm{Lo}_v \le F_v(p) = Bp + Aq \le \mathrm{Hi}_v$ because $p \in [a_0, b_0]$, and $i_{\mathrm{full}} \le W_\tau - 1$. □

4.2. Exact collision modulo p; the clean band.

Lemma 4.4 (mod p)**.** *If τ covers p, then $G_{\tau,j^*} \equiv \gamma^{i^*} \pmod p$. No hypothesis on any multiplicative order is needed.*

Proof. $B + AN - (Bp + Aq) = (p-1)(Aq - B)$, so $B + AN - c = \ell^2 g + (p-1)(Aq - B)$. Since $2^{p-1} \equiv 1 \pmod p$, $u^{B+AN-c} \equiv 2^{Lg} = \gamma^g \pmod p$. □

Lemma 4.5 (mod q)**.** *If τ covers p, then $G_{\tau,j^*} \equiv \gamma^{i^*} u^{(A-B)(p-1)} \pmod q$.*

Proof. Modulo $q-1$ we have $N \equiv p$. So $B + AN - c \equiv B + Ap - c$, while $\ell^2 g \equiv Bp + A - c$, and the difference is $(A-B)(p-1)$. □

Definition 4.6. The true giant G is *FULL* if $G \equiv \gamma^{i'} \pmod q$ for some $i' \in [0, m)$.

Lemma 4.7 (one-sided extraction)**.** *If τ covers p and its true giant G is not FULL, then* $\gcd(f(G), N) = p$.

Proof. The factor $i = i^*$ of $f(G) = \prod_{i<m}(G - \gamma^i)$ vanishes modulo p. Modulo q no factor vanishes, and $\mathbb{Z}/q\mathbb{Z}$ is a field. □

Lemma 4.8 (clean band)**.** *Call block k* clean for q *if no $\ell \in \mathcal{P}_k$ divides $q-1$. At least one of the four blocks is clean for q.*

Proof. Otherwise $q-1$ has four distinct prime factors exceeding Y, so $q-1 > Y^4 > \sqrt{2}P > q$. □

Lemma 4.9 (FULL on a clean block)**.** *If block k is clean for q, then the true giant is FULL if and only if $d \mid (A-B)(p-1) - \ell^2\delta$ for some $\delta \in [-i^*, m-1-i^*] \subset (-m, m)$.*

Proof. FULL holds iff $2^{(L/\ell^2)[(A-B)(p-1)-\ell^2\delta]} \equiv 1 \pmod q$ for some $\delta = i' - i^*$. Since $d \mid q-1$ and $\gcd(L, q-1) = 1$, the factor L/ℓ^2 cancels. □

Only the enlarged range $|\delta| < m \le \bar{m}_k$ is used for counting below.

4.3. **Counting lemmas.**

Lemma 4.10 (small orders are rare)**.** *For $Q_0 \ge 2$ and $T \ge 1$, $\#\{q > Q_0 \text{ prime} : \operatorname{ord}_q(2) \le T\} \le T(T+1)/(2\log_2 Q_0)$.*

Proof. Such a q divides $2^j - 1$ for some $j \le T$, and $2^j - 1$ has fewer than $j/\log_2 Q_0$ prime factors exceeding Q_0. □

Lemma 4.11 (coverage by the large sieve)**.** *Fix a prime $q > 16Y$ and a block k. Let $U_k(q)$ be the set of primes $p \in (P/2, P]$ with $\ell \nmid Bp - Aq$ for all $(v, \ell) \in V \times \mathcal{P}_k$. Then $|U_k(q)| \le CP/H^2$ for an absolute C.*

Proof.

- *The forbidden classes.* For $\ell \in \mathcal{P}_k$ we have $\ell \nmid qB$, so $\ell \mid Bp - Aq$ iff $p \equiv AqB^{-1} \pmod \ell$. These $|V|$ classes are distinct, because $0 < |AB' - A'B| < H^2 < \ell$. So $U = U_k(q)$ avoids $|V| < \ell$ classes modulo every $\ell \in \mathcal{P}_k$. Let $Q = \lfloor P^{1/2} \rfloor$ and $Z = \lfloor Q/2^{k+1}Y \rfloor$. Then $1 \le Z < Y$ and $\ln Z \ge c''n$ by (1). U also avoids the class 0 modulo every prime $r \le Z$, since $p > P/2 > Z$.
- *The large sieve.* Montgomery's arithmetic large sieve [19], with the constant of [20] (for a set avoiding $\omega(r)$ classes modulo each prime $r \le Q$), gives $|U| \le (P/2 + 1 + Q^2)/\mathcal{J}$, where $\mathcal{J} = \sum_{d \le Q} \mu^2(d) \prod_{r \mid d} \omega(r)/(r - \omega(r))$.
- *Bounding $\mathcal{J}$.* Keep only the terms $d = \ell a$ with $\ell \in \mathcal{P}_k$ and $a \le Z$ squarefree. These are squarefree and at most Q, because the primes of a are below $Y < \ell$. Each contributes $\frac{|V|}{\ell - |V|} \cdot \frac{1}{\varphi(a)}$. Using $\sum_{a \le Z} \mu^2(a)/\varphi(a) = \sum_{\mathrm{rad}(m) \le Z} 1/m \ge \sum_{m \le Z} 1/m \ge \ln Z$ (expand $1/\varphi(a) = \prod_{r \mid a}(1/r + 1/r^2 + \cdots)$), we get $\mathcal{J} \ge \lambda_k \ln Z$, where $\lambda_k := |V| \sum_{\ell \in \mathcal{P}_k} 1/\ell$.
- *Conclusion.* By the prime number theorem in the progression $3 \bmod 4$, $\sum_{\ell \in \mathcal{P}_k} 1/\ell \ge c'/n$, so $\lambda_k \ge c' H^2/(2n)$. Hence $|U| \le 2P/(\lambda_k c'' n) \le CP/H^2$.

□

Using only the prime moduli ℓ, as in the first version of this work, gives CnP/H^2 instead. The refinement by the presieve over $r \le Z$ was pointed out by a GPT-6.1-sol referee.

Lemma 4.12 (FULL is rare)**.** *Fix q with $d = \mathrm{ord}_q(2) > T$ and a block k clean for q. Let $F_k(q)$ be the set of integers $p \in [P/2, P]$ for which some $(v, \ell) \in V \times \mathcal{P}_k$ has $\ell \mid Bp - Aq$ and $d \mid (A-B)(p-1) - \ell^2\delta$ for some $|\delta| < \bar{m}_k$. Then*

$$|F_k(q)| \le 2|V||\mathcal{P}_k|\,\bar{m}_k\Big(\frac{PH}{2^{k+1}YT} + 1\Big).$$

Proof. Fix (v, ℓ, δ) and let $g' = \gcd(A-B, d) \le H$. The congruence on p is unsolvable or holds on one class modulo d/g'. Cleanliness gives $\ell \nmid d$. With $p \equiv AqB^{-1} \pmod{\ell}$, the CRT gives one class modulo $\ell d/g' \ge \ell T/H$. There are at most $|V||\mathcal{P}_k|(2\bar{m}_k - 1)$ triples. □

4.4. **One run.** For $T \ge 1$ (an analysis-only parameter) and $(p, q) \in \mathcal{S}_n$, let $k(q)$ be the least block clean for q. Define

$$E_a(T) = \{d \le T\}, \quad E_b(H) = \{d > T,\ p \in U_{k(q)}(q)\}, \quad E_c(H, T) = \{d > T,\ p \in F_{k(q)}(q)\}.$$

(The set E_b also depends on T; its bound does not, so we suppress T.)

Proposition 4.13 (one run; proved)**.** *There are absolute constants C, c such that for H in the range (1) and $1 \le T \le P/2$ the following hold.*

(i) $\mathsf{DCL}_{n,H}$ outputs a proper divisor of N or `FAIL` *on every input.*
(ii) It costs at most $Cn^c H^{5/3} 2^{n/6}$ bit operations on every n-bit input.
(iii) Every `FAIL` *pair of $\mathcal{S}_n$ lies in $E_a(T) \cup E_b(H) \cup E_c(H, T)$, and*

$$\frac{|E_a|}{|\mathcal{S}_n|} \le \frac{CnT^2}{P}, \quad \frac{|E_b|}{|\mathcal{S}_n|} \le \frac{Cn}{H^2}, \quad \frac{|E_c|}{|\mathcal{S}_n|} \le Cn\Big(\frac{H^{14/3}P^{1/3}}{T} + \frac{H^{10/3}}{P^{1/3}}\Big).$$

Proof.
- *Success certificate.* Suppose $d > T$, $p \notin U_{k(q)}(q)$ and $p \notin F_{k(q)}(q)$. By Lemma 4.2 some target of block $k(q)$ covers p, and by Lemma 4.3 its true giant is computed. By Lemma 4.9 it is not FULL, since the exact δ-range lies in $|\delta| < \bar{m}_k$. By Lemma 4.7 its gcd is p. So some block outputs a proper divisor.
- *Normalisation.* There are at most $3P/n$ primes $q \in (P/\sqrt{2}, \sqrt{2}P)$, and we divide by $|\mathcal{S}_n| \ge c_0 P^2/n^2$. This needs no prime counts in short intervals.
- E_a. By Lemma 4.10 with $Q_0 = 2^{(n-1)/2}$, at most $T(T+1)/(n-1)$ primes q qualify, each with at most $P/2 + 1$ partners.
- E_b. This follows from Lemma 4.11.
- E_c. Use $|V| \le H^2$, $|\mathcal{P}_k| \le 2^k Y$ and $\bar{m}_k \le 3HP/(4^k Y^2)$ in Lemma 4.12. This gives $|F_k(q)| \le 6H^4P^2/(Y^2T) + 6H^3P/Y$. Then sum over q and use $Y \ge P^{1/3}H^{-1/3}$.
- *(i).* Only validated gcds and band primes are output.
- *(ii).* See §4.5.

□

Remark 4.14 (Theorem D′)**.** Take $H = \lceil n2^{\varepsilon n/2} \rceil$ and $T = \lfloor 2^{(1/4-\varepsilon)n} \rfloor$. Then Proposition 4.13 gives cost $N^{1/6+5\varepsilon/6+o(1)}$ and exceptions $O(N^{-\varepsilon})$ for every fixed $\varepsilon < 1/52$. The binding term is the first E_c term, which needs $-1/12 + 13\varepsilon/3 < 0$. This is Theorem D′ of the project notes, stated there for $\varepsilon \le 1/60$.

A dyadic recount of E_c over the size of $\mathrm{ord}_q(2)$ removes E_a and certifies the unchanged algorithm for every $\varepsilon < 1/40$. Combining D′ at $\varepsilon = 1/55$ with a single Harvey–Hittmeir fallback gives average cost $N^{2/11+o(1)}$ on $\mathcal{S}_n$. This was observed by a GPT-6.1-sol referee and is superseded by Theorem 1.1. (The dyadic recount and the 2/11 claim are proved in `notes/avgtime_results.md`, Lemma D and §4. Neither is used below.)

4.5. **Bit complexity.** Per block, with T_k targets and $m \le \bar{m}_k$, the costs are as follows.

step	operations	bit cost
prime sieve to $16Y$; V; $\mathrm{Lo}_v, \mathrm{Hi}_v$	$O(Y \log\log Y)$; $O(H^2)$ square roots	$\tilde{O}(Y + H^2 n)$
integer product tree of $\{\ell^2\}$ ($\log_2 L_k \le 3 \cdot 2^k Y$)	balanced tree	$O(M(Y) \log Y)$
exponent tree; γ, γ^{-1}	$O(Y \log Y)$ mults mod N, 1 inversion	$O(Y \log Y\, M(n))$
roots, s, c, windows	$O(\lvert V\rvert\lvert\mathcal{P}_k\rvert \log Y)$ mults mod ℓ, ℓ^2	$\tilde{O}(H^2 Y)$
$w_\ell = u_\ell^N$; giants	$O(n\lvert\mathcal{P}_k\rvert + nT_k + T')$ mults mod N	$O((n(T_k+Y)+T')M(n))$
choice of m; small inversions	$O(nT_k)$ small-integer ops	$\tilde{O}(T_k n^{O(1)})$
babies γ^i	m mults	$O(mM(n))$
f by product tree; remainder tree at T' giants	$O((M(m) + M(T'))\log(m + T'))$ ring ops	$\tilde{O}((m+T')n)$
gcds	T' gcds of n-bit integers	$O(T' M(n) \log n)$

- Polynomial products over $\mathbb{Z}/N\mathbb{Z}$ use Kronecker substitution with slots wider than $2n + \lceil \log_2(D+1) \rceil$ [27].
- Remainders by monic polynomials use reversal and Newton inversion of a power series with constant term 1. So no inverse of a nonunit is ever needed.
- By step 5, $m + T' \le \bar{m}_k + T_k \le \frac{5}{2} HP/Y^2 + 4 + 2H^2 \cdot 2^{k+1} Y$.
- Summing over the four blocks gives $O(n^{O(1)}(HP/Y^2 + H^2 Y)) = O(n^{O(1)} H^{5/3} P^{1/3})$, since $Y \asymp P^{1/3} H^{-1/3}$. This proves Proposition 4.13(ii).

5. A fourth-moment coverage bound

The large sieve bounds the uncovered fraction by $O(1/\lambda)$, where $\lambda \asymp H^2/n$ is the expected number of covering events. Here we average over both p and q and control the joint statistics of up to four band primes. This gives $O(\lambda^{-2})$.

5.1. **Setting.**

- Let $X_\square \ge 1$ be an integer and $\mathfrak{B}_0 = (0, X_\square]^2 \cap \mathbb{Z}^2$.
- Let $\mathcal{P}$ be a nonempty set of primes in $(Y, 2Y]$, and V a nonempty set of coprime pairs $(A, B) \in [1, H]^2$.
- Put $s = \sum_{\ell \in \mathcal{P}} 1/\ell$ and $\lambda = |V| s$.
- For $x = (a, b)$ and $v = (A, B)$ put $F_v(x) = Ba - Ab$, and $z_\ell(x) = \#\{v \in V : \ell \mid F_v(x)\}$, $\mathcal{W}(x) = \#\{\ell \in \mathcal{P} : z_\ell(x) > 0\}$. The point x is *uncovered* if $\mathcal{W}(x) = 0$.
- Let $\mathcal{L} = \bigcup_{v \in V} \mathbb{R} v$ and $\mathfrak{B} = \mathfrak{B}_0 \setminus \mathcal{L}$. A prime pair $p \ne q$ with $p, q > H$ never lies on $\mathcal{L}$.
- A *configuration* of order j is $\mathbf{c} = ((\ell_1, v_1), \dots, (\ell_j, v_j))$ with distinct $\ell_i \in \mathcal{P}$ and $v_i \in V$. Its lattice is $\Lambda_{\mathbf{c}} = \{x : \ell_i \mid F_{v_i}(x)\ \forall i\}$, and $d_{\mathbf{c}} = \prod \ell_i$.

Theorem 5.1 (two-variable fourth moment; proved)**.** *Assume $H \ge 2$, $Y \ge 16H^2$, $HX_\square < Y^4$ and $\lambda \ge 1$. Then*

$$\#\{x \in \mathfrak{B} : \mathcal{W}(x) = 0\} \le X_\square^2 \Big(\frac{3}{\lambda^2} + \frac{1}{\lambda^3} \Big) + 2HX_\square + \lambda^{-4} \sum_{j=1}^{4} \kappa_j \mathrm{Err}_j,$$

with $\kappa_1 = 15\lambda^3$, $\kappa_2 = 25\lambda^2$, $\kappa_3 = 10\lambda$, $\kappa_4 = 1$ *and* $\mathrm{Err}_j = A_j + B_j + C_j + D_j + E_j + F_j$, *where:*

- $A_j = (|\mathcal{P}||V|)^j$;
- $B_j = 28.3\, X_\square (2Y)^{j/2} (2|V|)^j$;
- $C_j = 2HX_\square \sum_{r=\lceil j/2 \rceil}^{j} \binom{j}{r} \lambda^r |\mathcal{P}|^{j-r}$;
- $D_j = X_\square^2 |V| s^4$ *if* $j = 4$, *and* $D_j = 0$ *otherwise;*
- $E_j = 6HX_\square \sum_{r \in I_j} \binom{j}{r} \lambda^r |\mathcal{P}|^{j-r} + 2HX_\square |V| \sum_{r \in I_j} \binom{j}{r} s^{j-r} (|\mathcal{P}||V|)^r$, *with* $I_j = [\max(0, j-3), \lfloor j/2 \rfloor]$;
- $F_j = (3|V|)^j X_\square^2 s / Y$.

5.1.1. *Elementary facts.*

(F1) If $\ell \in \mathcal{P}$ does not divide both coordinates of x, then $z_\ell(x) \le 1$. Two events give $\ell \mid (AB' - A'B)a$ and $\ell \mid (AB' - A'B)b$, with $0 < |AB' - A'B| < H^2 < \ell$.

(F2) For $x \in \mathfrak{B}$, every $F_v(x)$ is a nonzero integer with $|F_v(x)| < HX_\square < Y^4$. So it has at most 3 prime factors in $(Y, 2Y]$, and $\sum_\ell z_\ell(x) \le 3|V|$.

(F3) $\Lambda_{\mathbf{c}}$ has index $d_{\mathbf{c}}$, by the CRT, since $x \mapsto F_{v_i}(x)$ maps onto $\mathbb{Z}/\ell_i$.

(F4) For $v \in V$ let $r_v(\mathbf{c}) = \#\{i : v_i \ne v\}$ and $t_v(\mathbf{c}) = \prod_{v_i \ne v} \ell_i$. Then $\Lambda_{\mathbf{c}} \cap \mathbb{R}v = \mathbb{Z} t_v v$, since $0 < |B_i A - A_i B| < \ell_i$ when $v_i \ne v$.

(F5) The shortest vector $u_{\mathbf{c}}$ of $\Lambda_{\mathbf{c}}$ has $|u_{\mathbf{c}}| \le 1.0746\sqrt{d_{\mathbf{c}}}$ (Hermite's constant in dimension two).

Lemma 5.2 (lattice points in a box)**.** *For a lattice* $\Lambda \subset \mathbb{R}^2$ *of covolume* d *with shortest vector length* ρ, *and a convex* K *of diameter* D, $|\#(\Lambda \cap K) - \mathrm{area}(K)/d| \le 1 + (4/\sqrt{3})D/\rho$.

Proof. Take a Lagrange–Gauss reduced basis b_1, b_2, so that $d \ge (\sqrt{3}/2)|b_1||b_2|$. In these coordinates K becomes a convex K' of area $\mathrm{area}(K)/d$. Its widths in the two coordinates are at most $(2/\sqrt{3})D/\rho$. Count $\mathbb{Z}^2 \cap K'$ column by column. Each column errs by at most 1, and the column lengths are unimodal, so the column sum errs by at most the other width. □

Lemma 5.3 (one configuration)**.** *Let* $\mathbf{c}$ *have order* $j \le 4$, $d = d_{\mathbf{c}}$ *and* $u = u_{\mathbf{c}}$.

(i) If $u \notin \mathcal{L}$, *then* $|\#(\Lambda_{\mathbf{c}} \cap \mathfrak{B}) - X_\square^2/d| \le 1 + 3.266 X_\square/|u| + \sum_{v'} X_\square/(t_{v'} \max(A', B'))$, *where the sum runs over* v' *with* $r_{v'}(\mathbf{c}) \ge \lceil j/2 \rceil$.

(ii) Otherwise $u = \pm t_v v$ *for one* $v \in V$. *Put* $r = r_v(\mathbf{c})$ *and* $g = d/t_v$; *then* $r \le \lfloor j/2 \rfloor$. *If* $j - r \ge 4$, *then* $\Lambda_{\mathbf{c}} \cap \mathfrak{B} = \emptyset$. *If* $j - r \le 3$, *then*

$$|\#(\Lambda_{\mathbf{c}} \cap \mathfrak{B}) - X_\square^2/d| \le \frac{2\sqrt{2}X_\square}{t_v|v|} + \frac{2HX_\square}{g} + \sum_{v' \ne v} \frac{X_\square}{t_{v'} \max(A', B')},$$

where again only v' *with* $r_{v'}(\mathbf{c}) \ge \lceil j/2 \rceil$ *occur.*

Proof. (a) *Directions* v' *not parallel to* u. The vector $t_{v'}v' \in \Lambda_{\mathbf{c}}$ is independent of u, so $|t_{v'}v'| \ge d/|u| \ge 0.93 Y^{j/2}$. On the other hand $|t_{v'}v'| \le (2Y)^{r_{v'}}\sqrt{2}H$. If $r_{v'} \le \lceil j/2 \rceil - 1$, this contradicts $Y \ge 16H^2$ for $j \le 4$. (The tightest case is $j = 3$, $r = 1$, which would need $\sqrt{Y} \le 3.3H$.) The points of $\Lambda_{\mathbf{c}} \cap \mathfrak{B}_0$ on $\mathbb{R}v'$ are $\sigma t_{v'}v'$ with $1 \le \sigma \le X_\square/(t_{v'} \max(A', B'))$.

(b) *Case (i).* Apply Lemma 5.2 to $\mathfrak{B}_0$ (diameter $\sqrt{2}X_\square$), and subtract the points on $\mathcal{L}$ counted in (a).

(c) *Case (ii),* $r \le \lfloor j/2 \rfloor$. We have $|u| = t_v|v| \ge Y^r$ and $|u| \le 1.0746(2Y)^{j/2}$. If $r > j/2$, then $Y^{1/2} \le 4.3$, a contradiction.

(d) *Case (ii), line structure.* For $x \in \Lambda_{\mathbf{c}}$, $\det(v, x) \in g\mathbb{Z}$. Each line $\{\det(v, \cdot) = mg\}$ meets $\Lambda_{\mathbf{c}}$ in a coset of $\mathbb{Z}u$. On $\mathfrak{B}_0$, $|\det(v, x)| < HX_\square$.

- If $j - r \ge 4$, then $g > Y^4 > HX_\square$, so only $m = 0$ occurs. That line is $\mathbb{R}v \subset \mathcal{L}$, which is removed.

- If $j-r\le 3$, let $h(\sigma)$ be the length of the chord $\{y\in[0,X_\square]^2:\det(v,y)=\sigma\}$. It is concave, $\max h\le\sqrt{2}X_\square$, and $\int h=|v|X_\square^2$. Summing $h(mg)/|u|\pm 1$ over the at most $2HX_\square/g$ nonzero m, and using $g|u|=d|v|$, gives the bound. Then subtract the other lines of $\mathcal{L}$ as in (a).

□

5.1.2. *Summation and assembly.* Summing Lemma 5.3 over all configurations of order j gives

$$\sum_{\mathbf{c}}\bigl|\#(\Lambda_{\mathbf{c}}\cap\mathfrak{B})-X_\square^2/d_{\mathbf{c}}\bigr|\le A_j+B_j+C_j+D_j+E_j.$$

The rows arise as follows.

- A_j. This is the "+1" term, summed over at most $(|\mathcal{P}||V|)^j$ configurations.
- B_j. In case (i) a configuration is counted at its own shortest vector u, which has $|u|\le R_j:=1.0746(2Y)^{j/2}$. Such a u has $|F_v(u)|\le 6.1HY^2<Y^3$, so it satisfies at most $2|V|$ events and lies in at most $(2|V|)^j$ lattices. Then use $\sum_{0<|u|\le R}1/|u|\le 8R$.
- C_j *and* E_j. Swap the sums over configurations and directions. For fixed v' and r, $\sum_{\mathbf{c}:r_{v'}=r}1/t_{v'}\le\binom{j}{r}\lambda^r|\mathcal{P}|^{j-r}$. Also $\sum_{v'}1/\max(A',B')<2H$ and $\sum_v 2\sqrt{2}/|v|\le 6H$. The $2HX_\square/g$ terms contribute $\sum_{\mathbf{c}:r_v=r}1/g\le\binom{j}{r}s^{j-r}(|\mathcal{P}||V|)^r$.
- D_j. Case (ii) with $j-r\ge 4$ occurs only for $j=4$, $r=0$, with error $X_\square^2/d$ summing to $X_\square^2|V|s^4$.

Put $F_j(x):=\sum_{\mathbf{c}}1[x\in\Lambda_{\mathbf{c}}]$. By (F1), $F_j(x)=(\mathcal{W}(x))_j$, the falling factorial, unless x lies in $\mathcal{D}=\{x:\exists\ell\in\mathcal{P},\ \ell\mid a,\ \ell\mid b\}$. On $\mathfrak{B}$, both quantities lie in $[0,(3|V|)^j]$ by (F2), and $\#\mathcal{D}\cap\mathfrak{B}_0\le X_\square^2 s/Y$. Hence

$$\Bigl|\sum_{\mathfrak{B}}(\mathcal{W})_j-X_\square^2\,\mathbb{E}(\mathcal{Z})_j\Bigr|\le\mathrm{Err}_j.$$

Here $\mathcal{Z}=\sum_\ell\mathcal{Z}_\ell$ with independent $\mathcal{Z}_\ell\sim\mathrm{Bernoulli}(|V|/\ell)$, so that $\mathbb{E}(\mathcal{Z})_j=\sum_{\mathbf{c}}1/d_{\mathbf{c}}$.

Now expand

$$(w-\lambda)^4=(w)_4+(6-4\lambda)(w)_3+(7-12\lambda+6\lambda^2)(w)_2+(1-4\lambda+6\lambda^2-4\lambda^3)(w)_1+\lambda^4.$$

For $\lambda\ge 1$ the coefficients are bounded by the κ_j. The same identity holds for $\mathcal{Z}$, and $\mathbb{E}(\mathcal{Z}-\lambda)^4\le\lambda+3\lambda^2$. Finally $\#\mathfrak{B}\le X_\square^2$, and each uncovered point contributes λ^4. So

$$\lambda^4\#\{\mathcal{W}=0\}\le X_\square^2(3\lambda^2+\lambda)+2HX_\square\lambda^4+\sum_j\kappa_j\mathrm{Err}_j,$$

which is Theorem 5.1. Every displayed inequality was also checked exhaustively on small boxes by code (`tests/test_sieve2d_bound.py`). □

5.2. **Consequences.** The explicit constants in Theorem 5.1 are poor. At the scale of Theorem 5.4 the error terms (dominated by B_4 and F_4) fall below the main term only for n of order 200–300. The bound is an asymptotic improvement over Lemma 4.11, not a statement about moderate sizes.

Theorem 5.4 (coverage exceptional set; proved)**.** *In the setting of Proposition 4.13, and with C absolute,*

$$\frac{|E_b(H)|}{|\mathcal{S}_n|}\le C\Bigl(\frac{n^4}{H^4}+n^6H^{1/3}N^{-1/6}\Bigr).$$

Proof. E_b is contained in the union over the blocks k of the pairs uncovered by block k. Prime pairs of $\mathcal{S}_n$ lie in $\mathfrak{B}$ with $X_\square=\lceil\sqrt{2}P\rceil$. The hypotheses of Theorem 5.1 hold for each block with $Y_k=2^kY$, by (2). We have $\lambda_k\ge c_1H^2/n$. If $\lambda_k<1$ then $H\ll\sqrt{n}$ and the claimed bound

exceeds 1, so it is trivial; assume $\lambda_k \geq 1$. With $|V| \asymp H^2$, $s \asymp 1/n$, $Y^3 H \asymp P$ and $H^2 \ll Y$, the whole weighted error sum of Theorem 5.1 satisfies

$$\frac{\lambda_k^{-4} \sum_j \kappa_j \mathrm{Err}_j}{X_\square^2} \ll H^{-6} + \frac{n^4 H^{-2/3}}{P^{1/3}} + \frac{n^3 H^{1/3}}{P^{1/3}}.$$

The three terms come from rows E (first part, $j = 4$), B and F; every other row is absorbed by these using the same relations (`src/sieve2d/bound.py` evaluates each row). So the uncovered integer points of block k number at most $X_\square^2[3\lambda_k^{-2} + \lambda_k^{-3} + O(H^{-6} + n^4 H^{1/3} P^{-1/3})]$. Multiplying by $n^2/(c_0 P^2)$, the population loss from integer to prime pairs, gives the bound. □

Proof of Theorem 1.2.

- Correctness and cost come from Proposition 4.13(i)–(ii).
- *(a).* Take $H = \lceil n 2^{\varepsilon n/4} \rceil$ and $T = \lfloor 2^{(1/4-\varepsilon/2)n}/n \rfloor$. Then $|E_a|/|\mathcal{S}_n| = O(nT^2/P) = O(2^{-\varepsilon n}/n)$. Theorem 5.4 gives $|E_b|/|\mathcal{S}_n| = O(2^{-\varepsilon n})$. The first E_c term has exponent $-1/12 + 5\varepsilon/3$, which is at most $-\varepsilon - 1/60$ for $\varepsilon \leq 1/40$. The second has exponent $-1/6 + 5\varepsilon/6$. The cost is $n^{O(1)} H^{5/3} 2^{n/6}$.
- *(b).* Take $H = \lceil n^{1+a} \rceil$ and $T = \lfloor 2^{n/5} \rfloor$. Then $E_a = O(n 2^{-n/10})$ and $E_c = n^{O(1)} 2^{-n/30}$. By Lemma 4.11 and Theorem 5.4, $E_b = O(\min(n^{-1-2a}, n^{-4a}))$.

□

5.3. **Where this method stops.** The following remark explains why the method stops at λ^{-2}.

Remark 5.5 (a moment barrier; proved for its stated scope)**.** Fix $j_0 \geq 2$, and let $\pi_\ell \in (0, 1/2]$ with $\max \pi_\ell \leq 2 \min \pi_\ell$ and $\sum \pi_\ell = \lambda \geq \lambda_{j_0}$. Peled, Yadin and Yehudayoff [22, Cor. 1.2(1), 1.3(1)] construct j_0-wise independent bits with a *common* marginal $\pi^* = \max \pi_\ell$ that are all 0 with probability $\geq c_{j_0} \lambda^{-\lfloor j_0/2 \rfloor}$ (for odd j_0 via their reduction (1.1)). Independent thinning by Bernoulli(π_ℓ/π^*) gives the marginals π_ℓ and keeps j_0-wise independence and the bound. Attach to each event a uniform label in V. Then all labelled event statistics of order at most j_0 agree exactly with the independent model, while the uncovered mass is $\geq c_{j_0} \lambda^{-\lfloor j_0/2 \rfloor}$. So no argument whose only input is the joint distribution of at most j_0 band-prime events can prove $o(\lambda^{-\lfloor j_0/2 \rfloor})$.

Our error analysis controls $j_0 \leq 4$; $j_0 = 5$ is sketched in the notes, and by the above it cannot improve on λ^{-2}. At $j_0 = 6$ the per-lattice error bounds add up to about $X_\square^2/H^2$, far above the target $X_\square^2 \lambda^{-3}$. This is a limit of this proof technique, not of the truth. Exploratory sixth moments agree with the independent model to within 4% (Section 10). A Poisson-quality bound $e^{-c\lambda}$ would need genuinely new arithmetic input. Any such proof must also exclude the integer dilates $t(H, H+1)$ with $\gcd(t, \prod_{\ell \in \mathcal{P}} \ell) = 1$, which are uncovered integer points. The barrier concerns limited-information arguments only; it is not an arithmetic impossibility theorem.

6. A complete algorithm with average cost $N^{1/6}$

6.1. **The small-order detector.**

Lemma 6.1 (detector; elementary)**.** *Let $N = pq$ be odd, $T \geq 1$, $m = \lceil \sqrt{T} \rceil$, $J' = \lceil T/m \rceil$, and $d_r = \mathrm{ord}_r(2)$. Form $f(X) = \prod_{0 \leq i < m} (X - 2^i)$ over $\mathbb{Z}/N\mathbb{Z}$, and compute $G_j = \gcd(f(2^{jm}), N)$ for $1 \leq j \leq J'$ by a product tree and a remainder tree. At the first j with $G_j > 1$, compute $\gcd(2^k - 1, N)$ for every $k \in ((j-1)m, jm]$, and output the first proper value; if there is none, the detector reports* `FAIL`*. If $\min(d_p, d_q) \leq T$ and $d_p \neq d_q$, this outputs the prime with the smaller order. The cost is $\tilde{O}(\sqrt{T}\, n)$ bit operations.*

Proof. Since $2^{jm} - 2^i = 2^i(2^{jm-i} - 1)$, $G_j > 1$ iff the block $((j-1)m, jm]$ contains a positive multiple of d_p or d_q. The first such block contains $d_{\min}$. At $k = d_{\min}$, the gcd is divisible by the prime of order $d_{\min}$ and not by the other, whose order does not divide $d_{\min}$. For smaller k in the block the gcd is 1. □

Lemma 6.2 (coincidences)**.** *Let* $C(T) = \{(p,q) \in \mathcal{S}_n : \operatorname{ord}_p 2 = \operatorname{ord}_q 2 \le T\}$. *Then* $|C(T)| \le C(T^3/P^2)|\mathcal{S}_n|$.

Proof. For each $k \le T$, $2^k - 1$ has fewer than $2k/(n-2)$ prime factors exceeding $P/2$, so at most $(2k/(n-2))^2$ pairs have both orders equal to k. Summing over k gives at most $4T^3/(n-2)^2$ pairs. □

6.2. **The algorithm LAD.** All parameters are n-only:

- $T = 2^{\lfloor 3n/10 \rfloor}$;
- $H_1 = \lceil \sqrt{n} \rceil$ and $H_{i+1} = \min(\lceil H_i^{6/5} \rceil, H_{\max})$, with $H_{\max} = n^2\, 2^{\lceil n/60 \rceil}$, stopping at $H_k = H_{\max}$.

Then $k = O(\log n)$ and $H_{i+1}^{5/3} \le 2^{5/3} H_i^2$. The constant n_0 is chosen so that (1) holds for every rung.

On input N:

0. If $n < n_0$, run Harvey–Hittmeir and stop. If $N < 4$ the output is trivial. If N is even, output 2 together with Harvey–Hittmeir's factorisation of $N/2$.
1. Run the detector (Lemma 6.1) with T.
2. For $i = 1, \dots, k$, run DCL_{n,H_i}.
3. If no step produced a divisor, run Harvey–Hittmeir [10], which outputs the complete factorisation.

Finish. As soon as a step outputs a proper divisor d, test d and N/d for primality with AKS [1]. If both are prime, output $\{d, N/d\}$; otherwise run Harvey–Hittmeir on N. On $\mathcal{S}_n$ the finish never calls Harvey–Hittmeir.

6.3. **Proof of Theorem 1.1.**

- *(i) Completeness.* Every step outputs only validated divisors, and otherwise Harvey–Hittmeir factors N completely.
- *(ii) Worst case.* Each rung costs at most $Cn^c H_i^{5/3} 2^{n/6} \le Cn^{c+10/3} 2^{7n/36}$, and there are $O(\log n)$ rungs. The detector costs $\tilde{O}(2^{3n/20})$, AKS is polylogarithmic, and Harvey–Hittmeir runs at most once.
- *(iii) Average: reach sets.* Let R_i be the pairs of $\mathcal{S}_n$ that reach rung i, and R_{HH} those that reach the fallback.
  - By Lemma 6.1, a pair passes the detector only if $d_q > T$ or it lies in $C(T)$.
  - By Proposition 4.13(iii), a pair with $d_q > T$ that fails rung $i-1$ lies in $E_b(H_{i-1}) \cup E_c(H_{i-1}, T)$.
  - Hence $R_i \subseteq C(T) \cup E_c(H_{i-1}, T) \cup E_b(H_{i-1})$ for $i \ge 2$, and similarly for R_{HH} with H_k. Let $W \le Cn^{16/5} 2^{n/5}$ be the worst-case total cost.
- *(iii) Average: the five contributions.*
  (a) The detector and rung 1, paid by every pair, cost $O(n^{c+5/6} 2^{n/6})$.
  (b) Rung $i \ge 2$ on $E_b(H_{i-1})$ contributes $Cn^c H_i^{5/3} 2^{n/6} \cdot \min(1, Cn/H_{i-1}^2) \le C' n^{c+1} 2^{n/6}$. Over the $O(\log n)$ rungs this totals $O(n^{c+1} \log n\, 2^{n/6})$.
  (c) Harvey–Hittmeir on $E_b(H_k)$ contributes $Cn^{16/5} 2^{n/5} \cdot n/H_{\max}^2 = O(n^{1/5} 2^{n/6})$.
  (d) Pairs in $C(T) \cup \bigcup_i E_c(H_i, T)$ pay at most W. By Lemma 6.2, $|C(T)|/|\mathcal{S}_n| = O(2^{-n/10})$. By Proposition 4.13(iii) with $H \le H_{\max}$, the E_c terms have exponents $7/90 + 1/6 - 3/10 = -1/18$ and $1/18 - 1/6 = -1/9$. So these pairs contribute at most $W n^{O(1)} 2^{-n/18} = n^{O(1)} 2^{13n/90} = o(2^{n/6})$.
  (e) The primality tests are polylogarithmic.
- *(iv)* is immediate, since the cost is a nonnegative function of N.

□

Remark 6.3. (a) The ratio 6/5 is not special. What matters is that the proved tail exponent 2 in H exceeds the cost exponent 5/3. With a tail $(n/H^2)^j$, any $j > 5/6$ suffices after adjusting the schedule. A better tail, such as Theorem 5.4, therefore changes only polylogarithmic factors of the average. (Within these bounds the first rung that fails on fewer than half the pairs already needs $H \gtrsim \sqrt{n}$, so a factor $n^{c+5/6}$ remains.)

(b) The detector is needed within the proved bounds. Without it the E_c count is limited by primes with $\mathrm{ord}_q(2) \approx \sqrt{P}$, and the best ladder exponent we can certify is 17/100 (`notes/avgtime_results.md`, §5.2).

(c) The finish's primality tests and the Harvey–Hittmeir fallback are standard. Only rungs, detector and fallback contribute to the exponent.

7. The worst case reduces to coverage

7.1. **Setting.**

- A *weight* is a primitive pair (A, B) with $1 \le |A|, B \le H$. *Signed* weights allow $A < 0$, and the diagonal $(1, 1)$ is allowed.
- A *block* is a set of primes $\ell \equiv 3 \pmod 4$ with $H < \ell < \sqrt{N/2}$.
- The pair (p, q) is *covered* by (block, weights) if $\ell \mid Bp - Aq$ for some weight and some prime ℓ of the block.

7.2. **The algorithm DCL-wc.** The algorithm sees only N and the public parameters H, V, the blocks $\mathcal{P}_1, \dots, \mathcal{P}_K$, T and B_0. For block k let ℓ_k^- be its least prime and $\bar{m}_k = \lfloor 3HP/(\ell_k^-)^2 \rfloor + 2$; every window below has $W_\tau \le \bar{m}_k$, because $\mathrm{Hi} - \mathrm{Lo} \le (B + \sqrt{2}|A|)\sqrt{N} + O(H) \le 3HP$. The baby length $m_k \le \bar{m}_k$ is chosen by the endpoint-competitive rule of Section 3, and a block with no targets is skipped.

0. If $N < 4$, output `FAIL`. If N is even, output 2. If $N = s^2$, output s.
1. *Base selection.* For $b = 2, 3, \dots, B_0$:
   - check $\gcd(b, N)$;
   - by baby-step giant-step polynomial evaluation, find the least $r \le T_{\mathrm{scan}} = T + O(\sqrt{T})$ with $\gcd(b^r - 1, N) > 1$;
   - if that gcd is proper, output it; if no such r exists, *select* b and go to step 2; otherwise record r_b.

   If every $b \le B_0$ was recorded, put $\Lambda = \mathrm{lcm}_b\, r_b$ and test $\gcd(1 + j\Lambda, N)$ for $1 \le j \le (\lfloor \sqrt{N} \rfloor - 1)/\Lambda$.
2. For each block, with $L = \prod \ell^2$, $u_\ell = b^{L/\ell^2}$ (by the exponent tree) and $\gamma = b^L$:
   - build the targets of Section 3 for all weights in V. For $A < 0$ use $\mathrm{Lo} = Ba_0 + \lfloor AN/a_0 \rfloor$ and $\mathrm{Hi} = Bb_0 + \lceil AN/b_0 \rceil$, since $x \mapsto Bx + AN/x$ is then increasing. Choose m as in step 5 there.
   - γ *pre-step.* Find the least $r \in [1, m)$ with $\gcd(\gamma^r - 1, N) > 1$, by a product tree and binary descent. If the gcd is proper, output it. If it equals N, run the *equal-order split*: $E = Lr$ is a multiple of $\mathrm{ord}_N(b)$ with known factorisation. For each prime power $\varrho^e \| E$ compute $y = b^{E/\varrho^e}$ and $\gcd(y^{\varrho^i} - 1, N)$ for $0 \le i \le e$, and output any proper value. Otherwise the common order D is now known; test $\gcd(1 + jD, N)$ for $1 \le j \le (\lfloor \sqrt{N} \rfloor - 1)/D$.
   - *Giants.* Evaluate $f = \prod_{i<m}(X - \gamma^i)$ at all giants as in Section 3 (for $A < 0$ this uses w_ℓ^{-1}, one inversion per prime). Output any proper gcd, and collect the giants with $\gcd(f(G), N) = N$.

- *Exploration.* For each such giant, descend the subproduct tree of f. At a node whose value has gcd N, evaluate both children. Output any proper gcd, and descend into the children whose gcd is N.
- *Exact leaf.* If leaf i is reached with $G \equiv \gamma^i \pmod N$, put $S = c + \ell^2(g_{\mathrm{lo}} + jm + i)$. If $S^2 - 4ABN = D_0^2$ is a perfect square, output a proper $\gcd((S \pm D_0)/2, N)$.

3. Output `FAIL`.

All outputs are validated.

7.3. **Theorem R.**

Theorem 7.1 (precise form of Theorem 1.3; proved)**.** *Let $N = pq \in \mathcal{S}_n$. Then:*

(i) DCL-wc outputs a proper divisor of N or `FAIL`;
(ii) if (p, q) is covered by $(\mathcal{P}_k, V)$ for some k, it outputs a proper divisor;
(iii) its cost is

$$\tilde{O}\Big(\sum_k \big(m_k + T'_k + |V||\mathcal{P}_k| + \textstyle\sum_{\ell\in\mathcal{P}_k} \log \ell\big) + B_0 T^{1/2} + \sqrt{N}/T + \sqrt{N}/\Psi(\sqrt{N/2} - 1, B_0)\Big)$$

ring operations modulo N, plus $\tilde{O}(|V||\mathcal{P}_k|)$ word operations. Here Ψ counts smooth numbers. With $T = \lceil N^{1/3} \rceil$ and $B_0 = \lceil n^4 \rceil$, the last three terms are $N^{1/6} n^{O(1)}$.

Lemma 7.2 (one-sided order dichotomy; after [11])**.** *For b coprime to N, write $o_p = \mathrm{ord}_p(b)$ and $o_q = \mathrm{ord}_q(b)$.*

(a) If $\min(o_p, o_q) \le T_{\mathrm{scan}}$, the scan returns $r_b = \min(o_p, o_q)$, and the gcd there is proper iff $o_p \neq o_q$.
(b) If every $b \le B_0$ is recorded, the Λ-enumeration finds p at cost at most $\sqrt{N}/\Psi(p-1, B_0)$.

Proof. (a) The first r with $\gcd(b^r - 1, N) > 1$ is $\min(o_p, o_q)$, and at $r = o_q < o_p$ the gcd is q.
(b) If $o_p = o_q = r_b$ for all $b \le B_0$, the subgroup $\langle 2, \dots, B_0\rangle \le \mathbb{F}_p^\times$ is cyclic of order Λ dividing $p-1$. It contains every B_0-smooth integer below p, so $\Lambda \ge \Psi(p-1, B_0)$. Then $p = 1 + j\Lambda$ with $j \le \sqrt{N}/\Lambda$. For $B_0 \ge (\ln p)^4$, the elementary bound $\Psi(x, (\log x)^4) \ge x^{3/4}/(\log x)^3$ gives $\sqrt{N}/\Lambda \le P^{1/4} n^3$. □

Lemma 7.3 (γ pre-step)**.** *One of the following holds:*

- *a proper divisor is output;*
- $\mathrm{ord}_p(\gamma) = \mathrm{ord}_q(\gamma) < m$*, and the equal-order split outputs a divisor, either directly or through the D-enumeration at cost at most $\sqrt{N}/T$;*
- $\mathrm{ord}_p(\gamma) \ge m$ *and* $\mathrm{ord}_q(\gamma) \ge m$.

Proof. Suppose $\gamma^o \equiv 1 \pmod N$. Then $o_p, o_q \mid E = Lo$; the factorisation of E is known from the band primes and from trial division of $o < m$, at cost $\tilde{O}(\sqrt{m})$. For each prime $\varrho \mid E$, $y = b^{E/\varrho^e}$ has order $\varrho^{v_\varrho(o_p)}$ modulo p and $\varrho^{v_\varrho(o_q)}$ modulo q. If these valuations differ for some ϱ, the corresponding gcd is p or q. Otherwise $o_p = o_q = D$, and $D > T$ because b was selected; since $D \mid p - 1$, the enumeration finds p. □

Lemma 7.4 (exploration)**.** *Assume* $\mathrm{ord}_p(\gamma), \mathrm{ord}_q(\gamma) \ge m$.

(a) Each giant has at most one $i < m$ with $G \equiv \gamma^i \pmod p$. So at most one node per tree level has gcd N, and the exploration of a giant costs $O(\log m)$ node evaluations.
(b) If the true giant G^* has $\gcd(f(G^*), N) = N$, its exploration outputs a proper divisor.

Proof. (a) The γ^i, $i < m$, are distinct modulo p.

(b) Every node on the path to leaf i^* has value divisible by p, so its gcd is p (which is output) or N. If leaf i^* is reached with $G^* \equiv \gamma^{i^*} \pmod N$, then $S = c + \ell^2 g = Bp + Aq$. Hence $S^2 - 4ABN = (Bp - Aq)^2$ and $\{(S \pm D_0)/2\} = \{Bp, Aq\}$. Finally $\gcd(Bp, N) = p$ because $1 \le B < q$. □

Proof of Theorem 7.1. (i) holds because all outputs are validated.

(ii) Steps 0–1 output a divisor or select b (Lemma 7.2). In a covering block, Lemma 7.3 either outputs a divisor or gives orders $\ge m$. The true target exists and its true giant satisfies $G^* \equiv \gamma^{i^*} \pmod p$, by Lemmas 4.2–4.4 (the sign of A plays no role). Then either $\gcd(f(G^*), N) = p$, or Lemma 7.4(b) applies. The fallback enumerations in earlier blocks are complete, so they cannot abort.

(iii) We account for each step in turn.

- *Base selection.* Each base costs $\tilde{O}(\sqrt{T})$ ring operations by baby-step giant-step polynomial evaluation. The Λ-enumeration runs at most once, at the cost of Lemma 7.2(b).
- *Exponent trees.* Per block, $u_\ell = b^{L/\ell^2}$ for all ℓ comes from the all-but-one exponent tree of §3, at $\tilde{O}(\sum_\ell \log \ell)$ ring operations.
- γ *pre-step.* This is a product tree of the $\gamma^r - 1$ and a binary descent, $\tilde{O}(m)$.
- *Equal-order split.* The powers b^{E/ϱ^e} for all prime powers $\varrho^e \| E$ come from one all-but-one exponent tree over these prime powers. The valuations v_ϱ are read off as the least i with $y^{\varrho^i} \equiv 1 \pmod N$, and $D = \prod \varrho^{v_\varrho}$. The D-enumeration runs at most once, because its branch ends with a factor.
- *Giants and remainder tree.* These cost $\tilde{O}(m + T')$, as in §4.5.
- *Exploration.* This is batched depth by depth. At each depth the node polynomials have total degree at most m, and by Lemma 7.4(a) at most two nodes per explored giant are evaluated, so at most $2T'$ evaluation points are assigned. Multipoint evaluation of each node at its assigned points therefore costs $\tilde{O}(m + T')$ per depth. There are $\lceil \log_2 m \rceil$ depths.
- *Target construction.* The term $|V||\mathcal{P}_k|$ charges the rejected (weight, prime) tests, at $\tilde{O}(1)$ word operations each.

□

The reference implementation (`src/dchworst/wcalg.py`) is a correctness prototype. Two steps there are unbatched (the γ pre-step scan and the exploration), so it does not meet the cost bound (iii). Its outputs are unaffected.

7.4. **Consequences.**

Hypothesis 7.5 ($\mathrm{DCH}_{\mathrm{pair}}(\kappa)$)**.** *For all large n and every $(p, q) \in \mathcal{S}_n$, some block $\mathcal{P}_k$ ($k = 0, \dots, 3$) of the design of $\mathsf{DCL}_{n,H}$ with $H = \lceil n^\kappa \rceil$ contains a prime ℓ with $\ell \mid Bp - Aq$, for some primitive (A, B) with $1 \le |A|, B \le H$. The diagonal $(1, 1)$ is allowed.*

Proof of Corollary 1.4.

- *(a).* Apply Theorem 7.1 with the design of $\mathsf{DCL}_{n,H}$ and signed weights. The list cost is $n^{O(1)} H^{5/3} 2^{n/6} = N^{1/6+o(1)}$.
- *(b).* Apply Theorem 7.1 with one block, the J primes, and all signed weights of height H. With $J = Y^j$ and $H = Y^{1/2} J^{-c}$ the list cost is $HP/Y^2 + JH^2$, which is balanced at $Y = N^{1/(5+2j(1-c))}$ with exponent $E(j, c) = (1 + j(1 - 2c))/(5 + 2j(1 - c))$. Since $\partial_j E$ has the sign of $3 - 8c$, for $c > 3/8$ the best choice is $j = 1$, which gives $Y = N^{1/(7-2c)}$ and $(1 - c)/(7/2 - c)$. The extraction overhead of Theorem 7.1 is $N^{1/6+o(1)}$, which does not exceed this.

□

Remark 7.6. (a) *Minkowski.* With one prime $\ell \asymp P^{2/5}$, signed weights of height $\sqrt{2\ell}$ and $(1,1)$, pigeonhole covers every pair. So DCL-wc recovers Harvey's unconditional worst-case exponent $1/5$. This is the case $c = 0$, and it is not new.

(b) *Theorem D′ sharpened.* With the design of $\mathsf{DCL}_{n,H}$, DCL-wc fails only on pairs uncovered by every block. That set has fraction $O(n/H^2)$ by Lemma 4.11 applied to block 0. The sets E_a and E_c disappear.

(c) *Squares.* An earlier draft stated a divisor-cover conjecture for every integer candidate $p \in [\sqrt{N/2}, \sqrt{N}]$ of every N coprime to 6, with $A \neq B$. That form is false. For $N = t^2$ with t prime and $p = t$, the quantity $Bp^2 - AN = (B - A)t^2$ has no band divisor, since $0 < |B - A| < H < \ell < t$. Hypothesis 7.5 quantifies over prime pairs only, and that is all Theorem 7.1 needs.

Proposition 7.7 ($\mathrm{DCH}_{\mathrm{pair}}$ needs $\kappa \geq 1/2$; proved)**.** *Let $\mathcal{L}$ be any set of primes in $(Y, 16Y]$ with $16Y \leq P^\theta$, $\theta < 1/2$, and let V be any weight set with $|V||\mathcal{L}| = o(P)$. Put $\lambda = |V| \sum_{\ell \in \mathcal{L}} 1/(\ell - 1)$. Then at most $(2/(1-\theta) + o(1))\lambda|\mathcal{S}_n| + o(|\mathcal{S}_n|)$ pairs are covered. For the four blocks of $\mathsf{DCL}_{n,H}$, a positive proportion of $\mathcal{S}_n$ is uncovered by every block when $H \leq 0.2\sqrt{n}$ (positive weights) or $H \leq 0.14\sqrt{n}$ (signed weights).*

Proof. For a prime p let I_p be the interval of q with $(p, q) \in \mathcal{S}_n$. Fix a condition (A, B, ℓ). The partners $q \in I_p$ with $\ell \mid Bp - Aq$ are the primes of one class modulo ℓ. Let $M_0 = |V||\mathcal{L}|$ be the number of conditions.

- *Interior p.* If $|I_p| \geq P/n^2$, the Brun–Titchmarsh inequality in the Montgomery–Vaughan form [20] bounds the partners by $2|I_p|/(\varphi(\ell)\log(|I_p|/\ell))$. Here $\log(|I_p|/\ell) \geq (1 - \theta - o(1))\log P$.
- *Endpoint p.* The p with $|I_p| < P/n^2$ lie within $O(P/n^2)$ of the ends of the p-range. For one condition they contribute $O\big((P/n^2)(P/(n^2\ell) + 1)\big)$ by the trivial count. Summed over all conditions this is $O(\lambda P^2/n^4 + M_0 P/n^2) = o(\lambda|\mathcal{S}_n|) + o(|\mathcal{S}_n|)$.
- *Summing.* Sum over interior p and over the conditions. The prime number theorem with de la Vallée Poussin's error term gives $\pi(x + y) - \pi(x) = (1 + o(1))y/\log x$ uniformly for $y \geq x/(\log x)^3$. So $|\mathcal{S}_n| = (1 + o(1))\sum_p |I_p|/\log P$, summed over interior primes p, and the interior contribution is at most $(2/(1-\theta) + o(1))\lambda|\mathcal{S}_n|$.
- *The constants.* For the four blocks, $\lambda \approx (6/\pi^2)H^2(\ln 16/2)/\ln Y \approx 7.3H^2/n$ for positive weights, and twice that for signed weights. With $\theta \approx 1/3$, so that the constant is 3, the covered fraction is below 1 when $H \leq 0.2\sqrt{n}$, respectively $H \leq 0.14\sqrt{n}$.

□

Remark 7.8 (barriers to proving coverage)**.**

- We have no design that provably covers every pair with any $c > 0$.
- Counting arguments with error budget $\gtrsim \sqrt{J}$ give only $c \leq 1/4$; this is a calculation, and it is heuristic as a barrier.
- Single-modulus pigeonhole gives $c = 0$.
- A proof that also covers coprime non-prime pairs would cover, for $x \asymp Y^{25/8}$, a structured set of about $Y^{1/4}$ integers $bx - a$ by band primes. We know of no method for sets this sparse.
- Exhaustive worst cases over all of $\mathcal{S}_{32}$ and $\mathcal{S}_{36}$ follow the Poisson extreme (Section 10).

8. A true-factor floor for local designs

This section is a theorem inside a cost model. It explains why $N^{1/6}$ is the natural limit of the method. It is not a lower bound for factoring.

8.1. **The model.**

Cost. For an input N, a design issues *targets* (labels) τ, each with a finite window $S_\tau \subset \mathbb{Z}$ of exponent values. A shared baby list, with any sumset structure, tests them. The design has cost X if $\#\{\tau\} \le X$ and $\sum_\tau |S_\tau| \le X^2$, either per input or in expectation over uniform n-bit N. A baby-step/giant-step search with list length $m \le X$ and at most X giants satisfies this, since every window costs $\lceil |S_\tau|/m \rceil \ge 1$ giants.

The model charges labels, not stored group elements: it assumes that one giant certifies at most one label. For the targets of this paper two labels share a giant only in negligible cases, but we have no general bound.

Generators. A *residue-local degree-one generator* ι consists of the following data:

- fixed integers A, B, not both 0;
- integers $M \ge 1$, K with $M \mid K$, and $D \ge 1$;
- a side, p or q;
- arbitrary maps $\kappa \mapsto R(\kappa) \subseteq \mathbb{Z}/M$ and $\kappa \mapsto S(\kappa) \subset \mathbb{Z}$ (finite sets), defined on keys $\kappa \in \mathbb{Z}/K$.

Write x for the class variable ($x = p$ on side p) and y for the partner. On input N, with $\kappa = N \bmod K$, the generator issues $|R(\kappa)|$ targets. They cover (p, q) iff

$$x \bmod M \in R(\kappa) \quad \text{and} \quad \lfloor (Bx + Ay)/D \rfloor \in S(\kappa). \tag{3}$$

Locality hypothesis (L1).

- $D/|A| \le C_1 M^2$ if $A \ne 0$, and $D/|B| \le C_1 M^2$ if $A = 0$;
- $K \le C_2 M^2$;
- $M, K \le P/2$.

Why this shape is general for degree one. In canonical normalisation, every identity-valid degree-one target has exponent $(Bx + Ay - c)/D$, with $D = C_d M^2$ and $c \in [0, D)$ fixed by the class and by $N \bmod D$ (the denominator lemma and its Lemma K′ in the multimod notes, §3.1(i)). For true pairs this exponent equals $\lfloor (Bx + Ay)/D \rfloor$, so any window placement is some finite set S. Moving a global offset into the window is harmless only when the offset is itself determined by the key; this is part of hypothesis (L1), not automatic. The count below never uses validity: it holds for every rule of shape (3). The generator family and its maps are fixed for each n; only the key varies with the input.

8.2. **The bound.**

Lemma 8.1 (fixed-key partner count)**.** *Fix x with $\gcd(x, K) = 1$ and a unit key κ, with $A \ne 0$. The partners $y \equiv \kappa x^{-1} \pmod K$ of x satisfying (3) number at most $\min\big(|S(\kappa)|(\lfloor D/(|A|K) \rfloor + 1),\ \lfloor \ell_y/K \rfloor + 1\big)$, where $\ell_y \le x$ is the length of the partner range.*

Proof. The partners form an arithmetic progression of step K. Each value $s \in S$ confines y to an interval of length $D/|A|$. □

Theorem 8.2 (T1; precise form of Theorem 1.5; proved)**.** *Assume (L1), and let a design be a union of generators. Suppose $\mathbb{E}_N \#\{\tau\} \le X$ and $\mathbb{E}_N \sum_\tau |S_\tau| \le X^2$, where $\mathbb{E}_N$ averages over uniform n-bit N. Then*

$$\Pr_{(p,q) \in \mathcal{S}_n} [\textit{the true pair is covered}] \le C\, n^2 X^{3/2} N^{-1/4},$$

with C depending only on C_1, C_2.

Proof. 1. *Sum by keys.* Both primes exceed K, so they are units modulo K, and each x has one partner class κx^{-1} per unit key κ. Let $\varphi(\kappa)$ denote the bound of Lemma 8.1. Then

$$\mathrm{Count}(\iota) \le \sum_{\kappa} \varphi(\kappa)\, \#\{x : x \bmod M \in R(\kappa)\},$$

and the last factor is at most $c_s|R(\kappa)|P/M$, with $c_s = 1$ on side p and $5/4$ on side q.

2. *Bound φ.* Using $\min(u,v) \le \sqrt{uv}$ and $K \le P/2$, $\varphi(\kappa) \le \sqrt{2P|S(\kappa)|(D/|A| + K)}/K$.
3. *Pass to $\mathbb{E}_N$.* Every residue class modulo K contains at least $2^{n-2}/K$ integers of $[2^{n-1}, 2^n)$. So $\sum_\kappa f(\kappa) \le 2K\, \mathbb{E}_N f(N \bmod K)$ for $f \ge 0$. With (L1), $\sqrt{D/|A| + K} \le M\sqrt{C_1 + C_2}$, which gives

$$\mathrm{Count}(\iota) \le c_1 P^{3/2}\, \mathbb{E}_N\big[|R(N \bmod K)|\, |S(N \bmod K)|^{1/2}\big].$$

4. $A = 0$. Swap sides: put $R'(\kappa) = \{\kappa r^{-1} \bmod M : r \in R(\kappa) \cap (\mathbb{Z}/M)^\times\}$ and $(A', B') = (B, 0)$. Discarding nonunit labels loses no true pairs, since $p, q > M$. The coverage of true pairs is identical, $|R'| \le |R|$, and $A' \ne 0$.
5. *Sum over the design and apply the budget.* Summing over generators gives $\mathbb{E}_N \sum_\tau |S_\tau|^{1/2}$, which is at most $\sqrt{\mathbb{E}\#\tau \cdot \mathbb{E}\sum |S_\tau|} \le X^{3/2}$ by Cauchy–Schwarz. Finally divide by $|\mathcal{S}_n| \ge c_0 P^2/n^2$.

□

The proof counts integer pairs and needs no flatness, height or prime equidistribution. Arbitrary finite windows are allowed: a sparse window counts by its cardinality, not its hull. This is what also covers searches over several rational bases with known multiplicative relations, after a reduction to prime-valuation coordinates (`notes/below16_results.md`, Prop. MB).

Proposition 8.3 (the search lies in the model; proved)**.** *$\mathsf{DCL}_{n,H}$ run with the baby-length rule $m = \bar m_k$ is dominated, at constant-factor cost, by a design satisfying (L1) with $M = K = \ell$ and $D = \ell^2$. (This rule suffices for Theorem 1.2.) By Theorem 8.2, any design in the model that succeeds on a constant fraction of $\mathcal{S}_n$ costs at least $N^{1/6}/(\log N)^{O(1)}$. So Theorem 1.2(b) is within a polylogarithmic factor of the best such design.*

Proof. The targets are $(A, B, \ell, \mathrm{root})$. The class set is the set of roots of $Br^2 \equiv AN$, whose key is $N \bmod \ell$. Replace each window by the n-only superset $[\lfloor F_{\min}/\ell^2 \rfloor - 1, \lfloor F_{\max}/\ell^2 \rfloor + 1]$ over all n-bit N. Its span is at most $2.12HP < \overline{\mathrm{span}}$, so each width is at most $\bar m_k + 1$. Then $\sum |S| \le \sum_k T_k(\bar m_k + 1) \le (\sum_k (\bar m_k + T_k))^2$. With $m = \bar m_k$ the list cost is $\sum_k (\bar m_k + T'_k) \ge \sum_k (\bar m_k + T_k)$, so the design has budget $X = O(\text{list cost})$. The coverage event of the design contains the success certificate of Proposition 4.13. □

Remark 8.4 (what the model does not cover)**.** The model leaves out several things.

- *Partial variants.* A second theorem T2 allows about $n/2$ bits of size information about N with interval windows and gives the same bound (`notes/below16_results.md` §2.2). Targets of degree ≥ 2 give $N^{1/5}$ on average, conditionally on a real-fibre hypothesis (ibid., Cor. F).
- *Excluded designs.* Not covered are keys finer than $C_2 M^2$, sparse windows combined with size information, several labels per giant, and global selectors that localise p cheaply.
- *Smooth-order methods.* These lie outside the model. Pollard's $p - 1$ method factors a positive proportion of the nearly balanced family $x^{1-\varepsilon} < p, q \le x$ (a family much larger than $\mathcal{S}_n$) at cost $N^{0.152}$ [28, Prop. 35.3], below the floor. By the same report (remarks after Prop. 35.3) it fails on a positive proportion for every smoothness bound up to $N^{0.33}$, so it gives no almost-all statement.

- *Relation methods.* A deterministic quadratic sieve would be heuristically subexponential on $\mathcal{S}_n$. Proving this needs smooth-value concentration and a nontrivial Jacobi character on the relation kernel, neither of which is known.

9. Unbalanced ratios

This section summarises `notes/unbalanced_results.md`, where the proofs are written out in full. An independent Opus critic and an adversarial GPT-6.1-sol referee reviewed them. Both accept Theorem 9.1, and both accept Theorem 9.2 after fixes, which were applied. A blind GPT-6.1-sol report supplied the construction of Theorem 9.2. Here we only sketch the proofs. Theorems 9.1 and 9.2 extend the large-sieve version D′ (Remark 4.14). The fourth-moment improvement of Theorem 1.2(a) is not carried over here.

9.1. **Ratio bands.** For $j \ge 0$ let

$$\mathcal{S}_{n,j} = \{(p,q) \text{ primes}: \ 2^j p < q \le 2^{j+1}p, \ 2^{n-1} \le pq < 2^n\},$$

and write $R = 2^j$. Then $p \asymp \sqrt{N/R}$ and $q \asymp \sqrt{NR}$. The prime number theorem gives $|\mathcal{S}_{n,j}| \ge c_0 2^n/n^2$ uniformly for $j \le n/3$. Exceptions are counted by pairs in each band. A bound $O(2^{-\varepsilon n})|\mathcal{S}_{n,j}|$ in every band $j \le \theta n$ therefore gives an $O(N^{-\varepsilon})$ fraction of the union $\mathcal{S}_n^\theta$, the pairs with $q/p \lesssim N^\theta$.

Theorem 9.1 (ratios up to N^ε; proved)**.** *Fix $\varepsilon \in (0, 1/60]$, and let $H = \lceil n2^{\varepsilon n/2} \rceil$ and Y, $\mathcal{P}_k$ be as in Section 3. For $2^j \le H^2/16$ replace V by the rectangle*

$$V_j = \{(A,B) : 1 \le A \le H2^{-j/2}, \ 1 \le B \le H2^{j/2}, \ \gcd(A,B) = 1, \ A \ne B\},$$

with windows computed over the p-range of band j, and keep one baby list per block for all bands. For $n \ge n_0(\varepsilon)$ the resulting algorithm has three properties:

- *it outputs only proper divisors or* `FAIL`*;*
- *it costs $N^{1/6+5\varepsilon/6}(\log N)^{O(1)}$ on every input;*
- *it fails on $O(2^{-\varepsilon n})|\mathcal{S}_{n,j}|$ pairs of every band with $2^j \le H^2/16$, hence on an $O(N^{-\varepsilon})$ fraction of the pairs with $q/p \le N^\varepsilon$.*

Sketch.
- *Window span.* On band j, $Bx + AN/x \le H2^{j/2}\sqrt{N/R} + H2^{-j/2}\sqrt{2RN} \le (1+\sqrt{2})H\sqrt{N}$. So the window bound $\overline{\text{span}}$, the baby length $\bar{m}_k$ and the band primes are those of $\mathsf{DCL}_{n,H}$. Only the number of targets grows, by the number of bands, $O(n)$.
- *Weights.* $|AB' - A'B| \le H^2 < \ell$ and $|V_j| \gg H^2$, so the lock and the coverage criterion hold verbatim. So do the extraction lemmas, the clean band ($q \le H2^{n/2} < Y^4$) and the large sieve, applied to the p-interval of length $\asymp q/R$.
- *FULL count.* Lemma 4.12 holds with $\gcd(A-B,d) \le H^2$ in place of H. With $T = \lfloor 2^{(1/4-\varepsilon)n} \rfloor$ this costs one factor H. The first E_c exponent becomes $-1/12 + 23\varepsilon/6$, which is below $-\varepsilon$ for $\varepsilon < 1/58$, and so for $\varepsilon \le 1/60$.

□

Theorem 9.2 (polynomial ratios; proved)**.** *Fix $\varepsilon \in (0, 1/60]$ and $\theta \in (0, 1/5 - 3\varepsilon]$, and put $h = H^2$. For each band with $H^2/16 < 2^j \le 2^{\theta n}$, use the following data:*

- *the weights $(1, B)$ with $2 \le B \le h$;*
- *the band scale $Y_j = \lceil (2^{(n+j)/2}/h)^{1/3} \rceil$;*
- *four blocks of primes $\ell \equiv 3 \pmod 4$ in $(2^k Y_j, 2^{k+1}Y_j]$, each with its own L, γ and baby list.*

Run this after the algorithm of Theorem 9.1. For $n \geq n_0(\varepsilon, \theta)$ the combined algorithm outputs only proper divisors or `FAIL`*, and costs*

$$N^{1/6+\max(5\varepsilon/6,\ \theta/6+2\varepsilon/3)}(\log N)^{O(1)}$$

on every input. It fails on $O(2^{-\varepsilon n})|\mathcal{S}_{n,j}|$ pairs of every band $j \leq \theta n$.

Sketch.

- *Cost.* The window span on band j is at most $18Q_j$ with $Q_j = 2^{(n+j+1)/2}$. So a block's list costs $O(Q_j/Y_j^2 + hY_j) = O(h^{2/3}Q_j^{1/3})$, and the geometric sum over bands gives $h^{2/3}2^{(1+\theta)n/6}n^{O(1)}$.
- *E_a and E_b.* With $T_j \asymp Q_j^{1/2}N^{-\varepsilon}$, the set E_a is an $O(n2^{-2\varepsilon n})$ fraction of $\mathcal{S}_{n,j}$. The large sieve with the $h-1$ distinct classes qB^{-1} gives $E_b = O(n^2/h)$, using $Y_j^2 \leq \sqrt{N/R}$, which holds for $\theta \leq 1/5$.
- *E_c.* Here $\gcd(B-1, d) \leq h$, and the FULL count gives

  $$O\big(n(h^{8/3}Q_j^{-1/6}2^{\varepsilon n} + h^{4/3}R^{5/6}N^{-1/6} + h^{2/3}N^{(2\theta-1)/3})\big).$$

  The second term is $O(2^{-\varepsilon n})$ when $R \leq 2^{(1/5-14\varepsilon/5)n-O(\log n)}$. This is where $\theta < 1/5$ enters.

A fine-band variant, with weights $(1, B)$ for $B \approx R$, has the same exponent, up to a constant $2^{O(1/\varepsilon)}$ coming from its $O(1/\varepsilon)$ blocks. Its FULL count fixes (p, B, ℓ, δ) and counts q instead. □

9.2. **The envelope.** Pollard–Strassen finds every prime factor up to z deterministically in $z^{1/2}(\log N)^{O(1)}$ [23, 25, 2]. At ratio $q/p = N^\theta$ we have $p \leq N^{(1-\theta)/2}$, so it costs $N^{(1-\theta)/4}$ in the worst case. Combining it with Theorem 9.2, every band of ratio N^θ is factored, except for $O(N^{-\varepsilon})$ of its pairs, at cost $N^{e(\theta)+O(\varepsilon)}$, where

$$e(\theta) = \min\Big(\frac{1+\theta}{6},\ \frac{1-\theta}{4}\Big), \qquad 0 \leq \theta \leq 1/3.$$

The first branch is valid for $\theta \leq 1/5 - 3\varepsilon$, and the second for all θ. The envelope $e(\theta)$ exceeds $1/6$ for every $0 < \theta < 1/3$. It peaks at $e(1/5) = 1/5$, at $p \approx N^{2/5}$ and $q \approx N^{3/5}$. A whole range of ratios $q/p \leq N^{\theta_0}$ with $\theta_0 \geq 1/5$ therefore costs $N^{1/5+O(\varepsilon)}$, which is no better than Harvey's worst-case bound.

9.3. **A full-coverage barrier.** The peak at $\theta = 1/5$ is not an artefact of our designs, at least for full coverage in the following model.

Model $\mathfrak{L}$ (one-dimensional Lehman-linear collision searches). Fix N and a ratio band R_0. The candidates are the integers $p \in [x_1, x_2]$ with $R_0 \leq N/p^2 \leq 2R_0$. A search consists of:

- baby lists $\{\gamma_i^e : e \in S_i\}$, where $S_i \subset \mathbb{Z}$ is an *arbitrary* set of size m_i;
- giants G, each compared against one list $i(G)$ and carrying a target $(\alpha, \beta, c_0, M, r, g_0)$ with integers α, β, c_0, g_0 and $M \geq 1$, where

  $$(\alpha, \beta) \neq (0, 0).$$

  A target with $\alpha = \beta = 0$ does not depend on p; its collision is two-sided and can never be extracted.

The giant covers a candidate $p \equiv r \pmod M$ if $\lfloor g(p) \rfloor \in g_0 + S_{i(G)}$, where $g(p) = (\alpha p + \beta N/p + c_0)/M^2$. For the true factor g is an integer (a lock), so this is exactly a collision. For other candidates it is a relaxation, which only enlarges coverage. For $\beta = 0$ the class is refined to the sub-class on which g is integral; a true factor outside it cannot collide. Rational normalisations by $D \mid \beta M^2$ are allowed. The cost is $X = \sum_i m_i + \#\{\text{giants}\}$. Nothing is assumed about the group, the base, the moduli (prime, composite or CRT products) or the side that is extracted.

Proposition 9.3 (Proposition O′; proved in $\mathfrak{L}$; full coverage)**.** *(a) If $\beta \neq 0$, a giant whose list has size m covers at most $2m + 8\sqrt{mx_2/R_0}$ candidates.*

(b) If $\beta = 0$, it covers at most m.

(c) Hence a search in $\mathfrak{L}$ that covers a fraction f of the candidates has

$$X \geq \min\bigl(\sqrt{fx_2}/4,\ (f/64)^{2/3}(x_2R_0)^{1/3}\bigr) \asymp \min\bigl(\sqrt{fp},\ f^{2/3}q^{1/3}\bigr).$$

At $R_0 = N^{1/5}$ this is $\gg f^{2/3}N^{1/5}$, with any number of moduli.

Sketch.
- *(a).* Write $p = r + Mx$, and assume without loss of generality that $\beta > 0$. Then $g(x)$ has $g'' = 2\beta N/p^3$; the factor M^2 cancels. So g is convex with curvature $\kappa \geq 2|\beta|N/x_2^3$. On each side of the minimum, $g' \geq \sqrt{\kappa(g - \min g)}$. Hence the preimage of the i-th window value has length $O(1/\sqrt{\kappa i})$. Summing over the m values gives $2m + O(\sqrt{m/\kappa})$.
- *(b).* The map is linear with a nonzero integer slope on the refined class.
- *(c).* The total coverage is at most $2X^2 + 8\sqrt{x_2/R_0}\,X^{3/2}$.

□

Remark 9.4 (scope of Proposition 9.3)**.**
- *Full coverage only.* This is a *full-coverage* statement: it bounds how many candidates one search covers for a fixed N. It is not an average-case bound. An almost-all algorithm only needs the true factor, and it could beat the bound only if its covered sets favoured the true factor beyond their density.
- *Sharpness.* Explicit searches attain the bound inside $\mathfrak{L}$. The witness for the $q^{1/3}$ branch uses relaxed, non-lock coverage. Real algorithms reach $e(\theta) + O(\varepsilon)$ for almost all pairs: Theorem 9.2 on the $q^{1/3}$ branch, and Pollard–Strassen (worst case, outside $\mathfrak{L}$) on the $\sqrt{p}$ branch.
- *Outside the model.* Targets nonlinear in (p, q), multi-dimensional searches, lattice methods and polynomial evaluation are not covered. Accidental collisions are not counted. The cost counts window comparisons, so processing a structured family of windows implicitly is not excluded.
- *Prior art.* Ward–Maestro [28, Prop. 3.7] treat unbalanced factors $p \approx N^b$ in their oblivious covering model, which has no per-target locks. With materialised giants they obtain $T \gg N^{\max(1/5,(1-b)/3)}$. Since $b = (1 - \theta)/2$, their branch $(1 - b)/3$ is our $q^{1/3}$ branch $(1+\theta)/6$, and it binds for $\theta \geq 1/5$. The curvature argument is the same as theirs.

  The two models are not nested. Model $\mathfrak{L}$ allows giants restricted to residue classes modulo arbitrary M, which their oblivious families lack. That is why our bound falls to $(1 + \theta)/6$ below $\theta = 1/5$, where Theorem 9.2 attains it for almost all pairs. Conversely, their forms have both coefficients at least 1, so their model has no analogue of our $\beta = 0$ ($\sqrt{p}$) branch. For $\theta > 1/5$ their floor is therefore higher than ours.

9.4. **Which populations the results serve.**

- *Bounded and $N^{o(1)}$ ratios.* For each fixed ε, every band with $q/p \leq N^{\varepsilon}$ has exponent $1/6 + O(\varepsilon)$ with $O(N^{-\varepsilon})$ exceptions per band (Theorems 1.2 and 9.1). Letting $\varepsilon \to 0$ gives $1/6 + o(1)$ with $o(1)$ exceptions.
- *Both primes at least $N^{1/3}$, counted by pairs.* Then θ has a continuous law on $[0, 1/3]$, with $\Pr[\theta \leq t] = \log_2((1+t)/(1-t))$ asymptotically. At cost $N^{1/6+\eta}$ our methods handle only a fraction of about 30η of these pairs. Exponent $1/5 - \eta'$ fails only on a fraction of about $30\eta'$, but our methods give no exponent below $1/5$ for $1 - o(1)$ of them. We do not claim that no method can.

- *All semiprimes up to X.* Almost all have a factor below X^δ, so the question is trivial with $o(1)$ exceptions. With power-saving exceptions nothing below 1/5 follows from our methods, since a fraction $\asymp \eta/\log\log X$ of these semiprimes has θ within η of 1/5.

10. Experiments

Confirmatory experiments were pre-registered: predictions, decision rules, designs and metrics were committed before any run. Exploratory analyses are labelled as such. The algorithms see only N. The harness samples pairs uniformly from $\mathcal{S}_n$ and assigns outcome classes afterwards. Toy-size successes are correctness checks, not evidence for exponents.

10.1. The search DCL (pre-registration `05d2bf5`).

Designs.

- **C1** is the algorithm of Section 3 with a finite-size schedule: H minimal with block-0 multiplicity $\lambda_0 \geq 3$, so $H \asymp \sqrt{n}$. At this H the proved exceptional bounds exceed 1 and are vacuous. The experiments therefore test correctness, the lemmas and the cost exponent, not the proved rate.
- **C2** is C1 with $\lambda_0 \geq \ln(P/\ln P) + 2$ (heuristically full coverage), up to 64 bits.
- **C3** is C1 with all odd band primes, Tonelli–Shanks roots and a charged nonresidue search.
- **HH** is a calibrated Harvey–Hittmeir implementation.

Metric. The cost proxy E counts n-bit modular-multiplication equivalents, with polynomial work charged as $\sum(D+1)\lceil\log_2(D+2)\rceil$. E_{first} counts the blocks up to the first success.
Lemma checks (28–40 bits, 240 pairs).

- All 240 pairs were correct and factored.
- There were 0 violations of Lemmas 4.1–4.5.
- There were 0 violations of "gcd $= p$ iff not FULL" on 3002 clean covering targets.

Cost (40–80 bits).

n	H/Y	succ.	C1 E_{first}	HH E	C1/HH
40	9/81	32/32	$1.03 \cdot 10^5$	$2.12 \cdot 10^4$	4.87
48	9/206	32/32	$3.26 \cdot 10^5$	$7.57 \cdot 10^4$	4.31
56	10/503	32/32	$1.01 \cdot 10^6$	$2.96 \cdot 10^5$	3.40
64	11/1461	32/32	$4.08 \cdot 10^6$	$9.43 \cdot 10^5$	4.33
72	11/3097	32/32	$1.06 \cdot 10^7$	$3.43 \cdot 10^6$	3.10
80	11/7804	32/32	$2.76 \cdot 10^7$	$1.18 \cdot 10^7$	2.35

- Over 48–80 bits, the least-squares slopes of $\log_2$ of the median E_{first} are 0.203 (C1), 0.196 (C3), 0.227 (C2, 48–64 bits) and 0.226 (HH).
- There were 0 incorrect outputs in 656 runs.
- With heuristically full coverage (C2) there is no slope advantage.
- The slope difference partly reflects the H schedule, and slopes are not exponents.
- Over 48–96 bits (1000 pairs per size), uncovered and clean fractions match the Poisson and $1 - \sum 1/(\ell - 1)$ predictions, and FULL occurred on at most 0.045% of covering targets.

10.2. Coverage statistics (pre-registration of `sieve2d`).

- The design was 228 cells of 10^6 sampled pairs at 48–96 bits, plus 620 exhaustive cells at 20–30 bits.

- Uncovered counts matched the independent-residue model, with pooled ratio 1.00001 over 202 cells. The decay was exponential, with fitted slope -0.9995 in λ_{eff}. Prime and integer pairs agreed to 1.00004.
- One frozen prediction failed: the fourth moment for *integer* pairs, through a heavy-tailed atom at $x \equiv (0,0)$. Every prime cell passed.
- Exploratory sixth moments for prime pairs were within $[0.961, 1.038]$ of the model.
- These data support the conjectural $e^{-\lambda}$ regime. They do not test the proved rate.

10.3. **Worst-case coverage and Theorem 7.1 (pre-registration of `dchworst`).**

- The worst-case weight height over all $6.2 \cdot 10^6$ pairs of $\mathcal{S}_{32}$ and $7.8 \cdot 10^7$ pairs of $\mathcal{S}_{36}$ follows the Poisson extreme in all 8 fits. Three frozen predictions failed: one was mis-stated, one failed with the model's own expected number of exceedances, and one failed only on the side of better-than-random coverage.
- DCL-wc had 0 failures on covered pairs, exhaustively over $\mathcal{S}_{26}$ and $\mathcal{S}_{28}$ (about $6.5 \cdot 10^5$ pairs) and on 36 adversarial cells. These included equal orders of 2 modulo p and q, small $\operatorname{ord}_q(2)$ and FULL-forcing families. The algorithm of Section 3 failed on 153 of these covered pairs.

11. Limitations and open problems

- **Worst case.** By Theorem 1.3, what is missing is coverage alone. Prove Hypothesis 7.5 for some κ, or any covering theorem with $c > 3/8$. The first test case is a covering statement for the union of short intervals $\{bx - a\}$ with $x \asymp Y^{25/8}$ and height $Y^{1/8}$.
- **Exceptional rate.** The proved tails are n/H^2 and n^4/H^4. The data support $e^{-cH^2/n}$. Remark 5.5 shows that this needs information beyond five-prime statistics.
- **Below 1/6.** Theorem 1.5 rules this out for residue-local degree-one collision designs. Remark 8.4 lists the escapes that remain open: a global selector, keys finer than M^2, label multiplicity, and relation methods. Smooth-order methods already beat 1/6 on a positive proportion, but not almost everywhere.
- **Ratios.** Is there a deterministic algorithm that factors almost all pairs with $p \approx N^{2/5}$, $q \approx N^{3/5}$, that is $q/p \approx N^{1/5}$, in time $N^{1/5-\delta}$? Within model $\mathfrak{L}$ (comparisons counted, accidental collisions ignored), Proposition 9.3 shows that this needs either a covering that favours the true factor beyond its density, or a mechanism outside one-dimensional Lehman-linear searches.
- **Practice.** At 40–80 bits the search is 2.3–4.9 times slower than Harvey–Hittmeir in our proxy. The heuristic subexponential methods (ECM, the quadratic and number field sieves) are far faster still. This work concerns rigorous deterministic bounds only.

Acknowledgements

This work was carried out with heavy AI assistance, and we describe the process so that readers can weigh it.

- *The human author* set the research question and direction, proposed several of the ideas tested, and decided which threads to pursue.
- *An orchestrating Claude session* (Anthropic, Claude Opus 5.5) planned the work and reviewed results.
- *Claude Opus 5.5 subagents* produced all code, proofs, experiments and notes. They worked under written rules: exact arithmetic, no access to the factors in operational code, status labels on every claim, and pre-registration before experiments. This paper was drafted by such a subagent from the project notes.

- *GPT-6.1-sol* (OpenAI, through the Codex CLI with web search) was used for literature sweeps, blind attempts and adversarial review.

Every theorem here passed at least one GPT-6.1-sol referee and one independent read-only Claude Opus critic, and each returned "accept with fixes". The fixes were applied. Some reviews found real errors:

- a base-order gap in a predecessor result (the shared base β^L can have tiny order);
- an implementation that returned N itself on some prime inputs;
- the false square-input form of the covering conjecture;
- the overclaim that the search is optimal in its model;
- a missing branch in an exponent envelope;
- a misquoted theorem number;
- an unprovable claim about the fifth moment.

No human has yet checked the proofs line by line. A Lean 4 formalisation covers several lemmas of Section 4, with the large sieve taken as an explicit hypothesis. Nothing here relies on it.

Appendix A. Full-candidate covering bounds (context; statements only)

This appendix is context only, and nothing in the main text depends on it. It concerns families that cover *every* integer candidate for a *fixed* N. It does not bound algorithms that only reach the true factor, and it is not a lower bound for factoring.

Priority. The exponent $1/6$, and the curvature argument behind it, are due to Ward–Maestro [28, Lemma 3.2, Cor. 3.3, Thm. 3.6], for Lehman forms with a common modulus. The same report gives the $N^{1/5}$ floor with materialised giants [28, Thm. 3.4] and bounded-degree forms [28, Thm. 9.4]. The variant below allows a separate lock modulus for each target. We obtained it independently and later. Its proofs are in the project notes (`notes/multimod_results.md` §3 and `notes/cycloharvey_results.md` §3), where they were checked by one AI critic and one AI referee. We do not reproduce them here.

Model. Fix $d_0 \geq 1$ and $\theta > 1$, and write $R = \sqrt{N}$. The candidates are the integers $p \in [R/\sqrt{\theta}, R]$. A target consists of:

- a modulus M with $\gcd(M, N) = 1$ and a unit class $r \bmod M$;
- an exponent map $h(x) = \sum_{i \in I} \alpha_i x^i - c$, read with $x^{-j} := (N/x)^j$, where $I \subseteq [-d_0, d_0] \setminus \{0\}$, $\alpha_i \in \mathbb{Z}[1/M] \setminus \{0\}$, and h is ℓ-integral on the whole class for every $\ell \mid M$;
- an integer window of width W.

A target covers p if $p \equiv r \pmod{M}$ and $h(p)$ lies in its window. The cost is X if there are at most X distinct triples (M, r, α), at most X targets, and $\sum W \leq X^2$. The targets of $\mathsf{DCL}_{n,H}$ fit this model with $M = \ell$ and $h = (Bp + Aq - c)/\ell^2$.

Theorem A.1 (full-candidate floors; proved in the project notes, not reproduced)**.** *In this model, constants depending on d_0 and θ only:*

(a) every covering family has $X \geq cN^{1/6}$;
(b) if degree-one targets cover fewer than half of the candidates, then $X \geq cN^{1/5}$;
(c) if the degree-one two-term targets use J distinct moduli, then $X \geq cN^{1/5}(JN^{o(1)})^{-1/5}$.

The integrality hypothesis bounds the denominators by $C_{d_0} M^K$. A degree-one two-term target then covers at most $C(RW/\eta)^{1/2} + C$ candidates, where η is its weight height, and M cancels. Part (a) follows by Cauchy–Schwarz. Part (c) also counts the admissible (class, weight) pairs per modulus. Lock lattices are multiplicative under the CRT, so covering *all* classes of a composite modulus gains nothing. What escapes the $N^{1/5}$ bound is *selective* covering, which is what $\mathsf{DCL}_{n,H}$ does. Whether it covers almost every candidate for a fixed N is not proved.

Appendix B. Reproducibility

Code, raw data and notes are in the project repository ([URL and archival revision withheld in this draft]). The main entry points are:

- `src/dcltheorem/` (the search);
- `src/avgtime/` (detector, ladder, exact exponent audit);
- `src/sieve2d/` (coverage bounds and experiments);
- `src/dchworst/` (the worst-case algorithm and coverage experiments);
- `src/below16/` (model checks).

Each has a test file under `tests/`. In the reference implementation of LAD, the fallback is Lehman's method and primality is a BPSW-type test. These are stand-ins for correctness tests at 32–40 bits. For example:

```
.venv/bin/python -m pytest -q tests/test_dcltheorem.py tests/test_avgtime.py \
    tests/test_sieve2d_bound.py tests/test_dchworst_wc.py tests/test_below16.py
PYTHONPATH=. .venv/bin/python -m src.dcltheorem.experiments lemma
```

References

[1] Manindra Agrawal, Neeraj Kayal, and Nitin Saxena. PRIMES is in P. *Annals of Mathematics*, 160(2):781–793, 2004.

[2] Alin Bostan, Pierrick Gaudry, and Éric Schost. Linear recurrences with polynomial coefficients and application to integer factorization and Cartier–Manin operator. *SIAM Journal on Computing*, 36(6):1777–1806, 2007.

[3] Daniel R. L. Brown. Lower bounds for straight line factoring. Cryptology ePrint Archive, Paper 2010/047, 2010.

[4] N. A. Carella. Deterministic integer factorization algorithms. arXiv:1308.2891v3, 2013. Latest version 22 April 2022.

[5] Edgar Costa and David Harvey. Faster deterministic integer factorization. *Mathematics of Computation*, 83(285):339–345, 2014. arXiv:1201.2116.

[6] Dragos Crisan and Radek Erban. On the counting function of semiprimes. arXiv:2006.16491v2, 2020.

[7] John B. Friedlander. Shifted primes without large prime factors. In *Number Theory and Applications (Banff, 1988)*, volume 265 of *NATO Adv. Sci. Inst. Ser. C Math. Phys. Sci.*, pages 393–401. Kluwer, 1989.

[8] Yiming Gao, Yansong Feng, Honggang Hu, and Yanbin Pan. On factoring and power divisor problems via rank-3 lattices and the second vector. arXiv:2512.19076v1, 2025.

[9] David Harvey. An exponent one-fifth algorithm for deterministic integer factorisation. *Mathematics of Computation*, 90:2937–2950, 2021. arXiv:2010.05450.

[10] David Harvey and Markus Hittmeir. A log-log speedup for exponent one-fifth deterministic integer factorisation. *Mathematics of Computation*, 91:1367–1379, 2022. arXiv:2105.11105.

[11] David Harvey and Markus Hittmeir. Deterministic methods for finding elements of large multiplicative order. arXiv:2601.11131v2, 2026.

[12] David Harvey and Joris van der Hoeven. Integer multiplication in time $O(n \log n)$. *Annals of Mathematics*, 193(2):563–617, 2021.

[13] Xinjie He and Amit Sahai. Refuting a conjecture of Umans and Wang on arithmetic-progression divisor covers. arXiv:2608.06681v1, 2026.

[14] Markus Hittmeir. A babystep-giantstep method for faster deterministic integer factorization. *Mathematics of Computation*, 87:2915–2935, 2018. arXiv:1608.08766.

[15] Markus Hittmeir. A time-space tradeoff for Lehman's deterministic integer factorization method. *Mathematics of Computation*, 90:1999–2010, 2021. arXiv:2006.16729.

[16] F. W. Lawrence. Factorisation of numbers. *Messenger of Mathematics*, 24:100–109, 1895.

[17] R. Sherman Lehman. Factoring large integers. *Mathematics of Computation*, 28(126):637–646, 1974.

[18] James McKee and Richard Pinch. Old and new deterministic factoring algorithms. In *Algorithmic Number Theory (ANTS-II)*, volume 1122 of *Lecture Notes in Computer Science*, pages 217–224. Springer, 1996.

[19] Hugh L. Montgomery. A note on the large sieve. *Journal of the London Mathematical Society*, 43:93–98, 1968.

[20] Hugh L. Montgomery and Robert C. Vaughan. The large sieve. *Mathematika*, 20:119–134, 1973.

[21] Ziv Oznovich and Ben Lee Volk. On deterministically finding an element of high order modulo a composite. arXiv:2506.07668v3, 2025.
[22] Ron Peled, Ariel Yadin, and Amir Yehudayoff. The maximal probability that k-wise independent bits are all 1. *Random Structures & Algorithms*, 38(4):502–525, 2011. arXiv:0801.0059.
[23] J. M. Pollard. Theorems on factorization and primality testing. *Proceedings of the Cambridge Philosophical Society*, 76:521–528, 1974.
[24] Pantelimon Stănică, Erik Mulder, and Markus Hittmeir. Multi-target hyperbolic sieves and elliptic trace obstructions. arXiv:2606.13018v2, 2026.
[25] Volker Strassen. Einige Resultate über Berechnungskomplexität. *Jahresbericht der Deutschen Mathematiker-Vereinigung*, 78:1–8, 1976/77.
[26] Chris Umans and Siki Wang. A number-theoretic conjecture implying faster algorithms for polynomial factorization and integer factorization. arXiv:2511.10851v1, 2025.
[27] Joachim von zur Gathen and Jürgen Gerhard. *Modern Computer Algebra*. Cambridge University Press, 3rd edition, 2013.
[28] Sean Ward and Maestro. Floors for deterministic integer factoring: covering models, partial information, the additive structure of the Lehman–Harvey start set, and three further floors. Technical report, iGent AI, September 2026. Technical report dated 2 September 2026; https://igent.ai/research/floors-not-bounds/factoring_report.pdf.